Department of Information Technology

ITA05 – COMPUTER VISION

ASSIGNMENT 1

Design, Compare and Justify a Computer Vision-Based Fabric Inspection System for Sri Lakshmi Textile Industries, Tirupur

CO1: Apply fundamental computer vision concepts, image acquisition, image formation and processing techniques to an industrial fabric-inspection application.

Student details	**Information**
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Course Code	ITA0508
Course Name	COMPUTER VISION

MARK SHEET – RUBRIC-BASED EVALUATION

Task	Title	Max marks	Awarded	Faculty remarks
1	Inspection approaches	4		
2	Pipeline design and operation	5		
3	Image representation	4		
4	Connectivity	4		
5	Image acquisition	5		
6	Sampling	5		
7	Quantization	4		
8	Image formation	5		
9	Implementation and performance	6		
10	Front-end adaptation	8		
	TOTAL	50		

Overall remarks

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MANDATORY SUBMISSION EVIDENCE CHECK

Task	Evidence expected	Included
1	Three-way comparison, limitations, justified choice	Yes
2	Three pipeline diagrams, stage I/O, complexity/time	Yes
3	Representation comparison and memory calculation	Yes
4	Neighbour diagrams and component analysis	Yes
5	Motion calculation and camera choice	Yes
6	Pixel counts, mm/px and defect sampling	Yes
7	Quantization levels and low-contrast analysis	Yes
8	Calibration, perspective and lighting design	Yes
9	System diagram, cost, storage, code and marker timing	Yes
10	Changed requirements and adaptation strategy	Yes

Evidence integrity note: calculations and design estimates are identified as such. No real factory measurements, vendor quotations or physical camera tests were supplied; the report does not present simulated or illustrative values as measured results.

INDUSTRIAL SCENARIO AND COMMON EXPERIMENTAL SETUP

Sri Lakshmi Textile Industries, Tiruppur, plans to inspect moving cotton and cotton-polyester fabric. The system must measure fabric width, detect holes, broken threads, oil stains, weaving irregularities and colour variations, estimate every defect location and command a downstream marking mechanism.

Parameter	Given specification	Design implication
Fabric width	1,000 mm	Full web coverage plus edge margins
Camera FOV	1,200 × 400 mm	Fabric occupies 83.3% of horizontal FOV
Conveyor speed	0.6 m/s	Motion blur and acquisition synchronisation
Smallest defect	1.5 mm	Several pixels should represent each defect
Oil contrast	5–8 grey levels	Stable illumination and adequate bit depth
Width accuracy	±2 mm	Geometric calibration at fabric plane
Decision time	0.5 s maximum	Capture-to-decision pipeline must be bounded
Camera-to-marker	0.3 m	Nominal travel time is 0.5 s
Lighting	500–3,000 lux	Control ambient light and flicker
Camera height	700 mm	Lens geometry and calibration required
Computer	4-core CPU, 16 GB RAM, no GPU	Optimised OpenCV/NumPy pipeline
Software	Python, OpenCV, NumPy	CPU-oriented implementation

Network	Gigabit Ethernet	Camera and PLC communication
Storage	2 TB	Seven-day retention constraint
Budget	₹1,50,000	Balanced procurement
Missed defects	Maximum 2%	Representative validation dataset

Common coordinate convention

Image x increases across the fabric from left to right. Image y increases in the direction of fabric travel. A calibrated transform converts pixel positions into millimetres. A defect record stores roll ID, timestamp, defect class, confidence, bounding box, centroid, encoder position, software version and marking status.

Marker timing calculation

Travel time = distance / speed = 0.3 m / 0.6 m/s = 0.5 s. This equals the entire stated decision limit, leaving no time for actuator delay or communication. Therefore, the practical software target should be below 0.35 s and the PLC should schedule the marker using encoder position rather than a fixed delay.

Assumptions

- The camera is mounted approximately normal to a flat, tensioned fabric plane.
- Lighting is enclosed and driven without visible flicker.
- The pixel-to-mm mapping is calibrated using a traceable grid at the fabric plane.
- Defect thresholds are tuned separately for fabric recipes and blends.
- Costs are planning estimates, not supplier quotations.
- The existing PC and storage are assumed to be available and excluded from the component budget.

SHARED PYTHON / OPENCV IMPLEMENTATION FOUNDATION

The same software foundation is reused across the tasks. It separates image preparation, candidate extraction, component measurement and decision logging. It is deliberately conventional and CPU-friendly.

```
# Install: python -m pip install opencv-python numpy matplotlib
import cv2
import numpy as np
import time
import json
def preprocess(frame):
    smooth = cv2.GaussianBlur(frame, (3, 3), 0.6)
    hsv = cv2.cvtColor(smooth, cv2.COLOR_BGR2HSV)
    gray = cv2.cvtColor(smooth, cv2.COLOR_BGR2GRAY)
    return smooth, hsv, gray
def flat_field(gray, reference):
    ref = np.maximum(reference.astype(np.float32), 1.0)
    img = gray.astype(np.float32)
    corrected = img * (np.median(ref) / ref)
    return np.clip(corrected, 0, 255).astype(np.uint8)


def candidate_masks(frame):
    bgr, hsv, gray = preprocess(frame)
    local = cv2.GaussianBlur(gray, (31, 31), 0)
    residual = cv2.absdiff(gray, local)
    contrast = (residual >= 5).astype(np.uint8) * 255
    sat = hsv[:, :, 1].astype(np.float32)
    sat_local = cv2.GaussianBlur(sat, (31, 31), 0)
    colour = (cv2.absdiff(sat, sat_local) > 12).astype(np.uint8) * 255
    return contrast, colour, gray, hsv
```

```
def components(mask, mmx, mmy):
    n, lab, stats, cent = cv2.connectedComponentsWithStats(
        (mask > 0).astype(np.uint8), connectivity=8)
    out = []
    for k in range(1, n):
        x,y,w,h,area = stats[k]
        cx,cy = cent[k]
        out.append({"bbox_mm":      [x*mmx,y*mmy,w*mmx,h*mmy],
                    "centroid_mm": [cx*mmx,cy*mmy],
                    "area_mm2": area*mmx*mmy})
    return out


def defect_record(kind, confidence, centroid, encoder_mm, time_s):
    return {"class":kind, "confidence":float(confidence),
            "centroid_mm":centroid,     "encoder_mm":float(encoder_mm),
            "timestamp_s":float(time_s),
            "action":"MARK" if confidence >= 0.70 else "REVIEW"}
```

The thresholds in this foundation are starting values only. A production system must add recipe-specific texture models, class decisions, calibration transforms, duplicate suppression, encoder scheduling and PLC fault handling.

TASK 1 – Overview and Features of Computer Vision – Compare Inspection Approaches [4 Marks]

Aim and industrial objective

The aim of Task 1 is to compare the candidate approaches using the actual textile production constraints. The selected technique must support a 1,000 mm moving web, 1.5 mm minimum defect, 5–8 grey-level oil-stain contrast, ±2 mm width accuracy, a 0.5 s decision limit and a four-core CPU.

Key condition	Value	Why it matters
Web speed	0.6 m/s	Motion and timing
Minimum defect	1.5 mm	Sampling and optics
Width tolerance	±2 mm	Calibration
Compute	4-core CPU	Algorithm complexity
Budget	₹1,50,000	Component choice

1. Working principles

Manual inspection uses trained workers to observe the moving fabric and identify defects by eye. Image-processing-assisted inspection captures images, enhances them and highlights suspicious regions, but an operator still confirms every decision. Fully automated computer vision acquires images, extracts features, classifies defects, measures width and sends a marker command without waiting for a human decision.

Approach	Strength	Limitation	Suitability
Candidate A	Simple / low cost	Limited context or performance	Low to moderate
Candidate B	Improves visibility or measurement	Needs calibration / extra processing	Moderate
Candidate C	More complete decision capability	Higher complexity or cost	Best when validated

2. Accuracy, speed and consistency

Manual accuracy depends on worker training, attention, fatigue, viewing angle and the visibility of a defect. Assisted inspection improves the visual presentation but retains human variability and response time. Automated inspection applies the same rules to every image and can operate continuously if the camera, CPU and marker interface meet the timing target. Its accuracy is not automatically perfect: it depends on representative training/validation data, calibration and lighting.

3. Labour, traceability and suitability

Manual inspection requires continuous staffing and usually provides limited digital traceability. Assisted inspection reduces visual strain and can retain images, but operators must review alerts. Automated inspection reduces routine labour and can save images, defect coordinates, timestamps and decisions. For a 0.6 m/s production line, automatic inspection is the most suitable primary method.

4. Limitations and mitigation

Automation can create false alarms from normal weave texture, miss low-contrast oil stains, fail on unfamiliar fabric recipes and drift after camera movement. Mitigation includes recipe-specific calibration, stable illumination, a review queue for low-confidence detections, periodic reference samples and human audits. A safe fallback must be defined for camera, lighting, PC or PLC failure.

5. Comparison table

Manual: flexible but slow and variable. Assisted: improved visibility but still human-limited. Automated: repeatable and traceable but requires engineering and validation. The recommended system is automated vision with human supervision rather than a fully unattended quality policy.

6. Industrial calculation and decision

At 0.6 m/s the web travels 600 mm each second. A 1.5 mm defect passes a fixed point in 1.5/600 = 0.0025 s, or 2.5 ms. Human reaction cannot reliably mark such a small feature at this speed. Camera-based acquisition and encoder position tracking are therefore required.

7. Task conclusion

I select fully automated computer vision as the primary inspection method. It best supports continuous inspection, ±2 mm width measurement, defect localisation and marking.

Task-specific decision

The final choice is the approach that provides sufficient defect visibility and localisation while remaining explainable, maintainable and compatible with the available PC and budget. The recommendation must be verified on real fabric samples before production acceptance.

Evidence to collect during commissioning

- Capture defect-free fabric and each defect type under the selected lighting.
- Record known defect dimensions and confirm their pixel representation.
- Measure processing latency on the actual four-core PC.
- Validate width at multiple positions against a traceable reference.
- Compare detected defects with a human-labelled ground-truth map.
- Record false alarms, missed defects, marker position error and storage consumption.

TASK 1 – CALCULATIONS, IMPLEMENTATION AND OBSERVATIONS

Calculation / implementation notes

The numerical decision must be tied to the scenario: fabric speed, sampling, contrast, calibration, CPU time and marker position. No unmeasured accuracy or timing value is claimed.

Relevant OpenCV / NumPy implementation

```
# Generic CPU-side processing pattern
import cv2
import numpy as np
import time


def process_one(frame):
    start = time.perf_counter()
    blur = cv2.GaussianBlur(frame, (3, 3), 0.6)
    gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)
    local = cv2.GaussianBlur(gray, (31, 31), 0)
    residual = cv2.absdiff(gray, local)
    mask = (residual >= 5).astype(np.uint8)
    n, labels, stats, centres = cv2.connectedComponentsWithStats(
        mask, connectivity=8)
    elapsed_ms = (time.perf_counter() - start) * 1000
    return stats[1:], centres[1:], elapsed_ms


# Actual elapsed_ms must be measured on the target PC and camera image.
```

Observation and interpretation

The code demonstrates the structure of an image-processing experiment. It does not provide a factory detection score because no labelled fabric image dataset was supplied. During implementation, the output should be compared with known defect coupons and recorded in the validation tables in the appendices.

Limitations

- Thresholds are fabric-dependent and require recipe calibration.

- The sample code is a candidate generator, not a fully validated defect classifier.
- The actual camera frame rate, image format and transfer overhead must be measured.
- A defect decision must be converted to encoder position before the marker can act.

TASK 2 – Levels of Computer Vision – Pipeline Design and Operation [5 Marks]

Aim and industrial objective

The aim of Task 2 is to compare the candidate approaches using the actual textile production constraints. The selected technique must support a 1,000 mm moving web, 1.5 mm minimum defect, 5–8 grey-level oil-stain contrast, ±2 mm width accuracy, a 0.5 s decision limit and a four-core CPU.

Key condition	Value	Why it matters
Web speed	0.6 m/s	Motion and timing
Minimum defect	1.5 mm	Sampling and optics
Width tolerance	±2 mm	Calibration
Compute	4-core CPU	Algorithm complexity
Budget	₹1,50,000	Component choice

1. Three candidate designs

Low-level processing maps pixels to improved pixels through denoising, colour conversion, filtering and thresholding. A three-level pipeline adds segmentation and feature extraction, then interprets those features to decide defect class and location. A deep-learning pipeline sends normalised images through a neural network that returns classes and boxes or masks.

Approach	Strength	Limitation	Suitability
Candidate A	Simple / low cost	Limited context or performance	Low to moderate
Candidate B	Improves visibility or measurement	Needs calibration / extra processing	Moderate
Candidate C	More complete decision capability	Higher complexity or cost	Best when validated

2. Pipeline diagrams

Low-level: Acquire → Denoise → Convert/Enhance → Threshold → Candidate mask. Three-level: Acquire → Preprocess → Segment → Extract texture/colour/shape → Classify → Measure →

Mark. Deep learning: Acquire → Normalise → CNN/detector inference → Class and mask/box → Coordinate conversion → Mark.

3. Stage inputs and outputs

Acquisition receives reflected light and outputs an image array. Preprocessing receives raw pixels and outputs corrected images. Segmentation outputs candidate regions. Feature extraction converts regions into measurable attributes such as area, aspect ratio, orientation, local variance and colour difference. Decision logic outputs class/confidence. Calibration and encoder mapping output physical coordinates.

4. Defect-specific stage requirements

Holes require contrast, edges, contours and area. Broken threads require directional filtering and continuity analysis. Oil stains require flat-field correction, local intensity/saturation differences and irregular-region analysis. Weaving irregularities require periodic texture, yarn spacing and orientation. Colour variation requires calibrated colour channels and comparison with a recipe reference.

5. Complexity and explainability

Low-level processing is fast but lacks context and can confuse shadows with stains. The three-level pipeline is moderately complex and explainable because each decision can be traced to a feature and rule. Deep learning can learn complex patterns but needs representative labelled images and may be too slow on a four-core CPU without a GPU.

6. CPU timing plan

A design target is 50 ms for acquisition/transfer, 60 ms for preprocessing, 120 ms for candidate extraction/features, 40 ms for classification/coordinate conversion, 20 ms for PLC communication and 60 ms reserve. Total target = 350 ms. These are engineering targets, not measured timings; benchmark on the actual PC.

7. Task conclusion

I recommend the three-level conventional pipeline. It has enough context to separate fabric defects from normal weave patterns and remains explainable and CPU-friendly. Deep learning can be evaluated offline or for selected difficult classes after CPU performance is proven.

Task-specific decision

The final choice is the approach that provides sufficient defect visibility and localisation while remaining explainable, maintainable and compatible with the available PC and budget. The recommendation must be verified on real fabric samples before production acceptance.

Evidence to collect during commissioning

- Capture defect-free fabric and each defect type under the selected lighting.
- Record known defect dimensions and confirm their pixel representation.
- Measure processing latency on the actual four-core PC.
- Validate width at multiple positions against a traceable reference.
- Compare detected defects with a human-labelled ground-truth map.
- Record false alarms, missed defects, marker position error and storage consumption.

TASK 2 – CALCULATIONS, IMPLEMENTATION AND OBSERVATIONS

Calculation / implementation notes

The numerical decision must be tied to the scenario: fabric speed, sampling, contrast, calibration, CPU time and marker position. No unmeasured accuracy or timing value is claimed.

Relevant OpenCV / NumPy implementation

```
# Generic CPU-side processing pattern
import cv2
import numpy as np
import time


def process_one(frame):
    start = time.perf_counter()
    blur = cv2.GaussianBlur(frame, (3, 3), 0.6)
    gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)
    local = cv2.GaussianBlur(gray, (31, 31), 0)
    residual = cv2.absdiff(gray, local)
    mask = (residual >= 5).astype(np.uint8)
    n, labels, stats, centres = cv2.connectedComponentsWithStats(
        mask, connectivity=8)
    elapsed_ms = (time.perf_counter() - start) * 1000
    return stats[1:], centres[1:], elapsed_ms


# Actual elapsed_ms must be measured on the target PC and camera image.
```

Observation and interpretation

The code demonstrates the structure of an image-processing experiment. It does not provide a factory detection score because no labelled fabric image dataset was supplied. During implementation, the output should be compared with known defect coupons and recorded in the validation tables in the appendices.

Limitations

- Thresholds are fabric-dependent and require recipe calibration.

- The sample code is a candidate generator, not a fully validated defect classifier.
- The actual camera frame rate, image format and transfer overhead must be measured.
- A defect decision must be converted to encoder position before the marker can act.

TASK 3 – Image Representation Analysis [4 Marks]

Aim and industrial objective

The aim of Task 3 is to compare the candidate approaches using the actual textile production constraints. The selected technique must support a 1,000 mm moving web, 1.5 mm minimum defect, 5–8 grey-level oil-stain contrast, ±2 mm width accuracy, a 0.5 s decision limit and a four-core CPU.

Key condition	Value	Why it matters
Web speed	0.6 m/s	Motion and timing
Minimum defect	1.5 mm	Sampling and optics
Width tolerance	±2 mm	Calibration
Compute	4-core CPU	Algorithm complexity
Budget	₹1,50,000	Component choice

1. RGB representation

RGB stores red, green and blue intensity values at each pixel. It preserves original colour and is important for colour variation. It is also useful when dyed warp and weft yarns have distinct colours. Brightness and colour are mixed in the channels, so illumination variation can affect all three values.

Approach	Strength	Limitation	Suitability
Candidate A	Simple / low cost	Limited context or performance	Low to moderate
Candidate B	Improves visibility or measurement	Needs calibration / extra processing	Moderate
Candidate C	More complete decision capability	Higher complexity or cost	Best when validated

2. HSV representation

HSV stores hue, saturation and value. Hue describes colour family, saturation describes colour strength and value represents brightness. Separating brightness from hue/saturation helps identify

colour changes and stains. Hue is unreliable in nearly grey or low-saturation pixels, so it should not be used alone.

3. Grayscale representation

Grayscale stores one intensity value per pixel. It is efficient for edge detection, texture analysis, local contrast and thread continuity. It cannot reliably distinguish two different colours that have similar brightness, so it is not sufficient for colour variation inspection.

4. Suitability by defect

Colour variation: RGB and HSV are strongest. Oil stain: HSV saturation/hue and grayscale local contrast complement each other. Hole: grayscale edge and contrast are useful. Broken thread: grayscale directional texture is useful, with colour added for dyed yarns. Weaving irregularity: grayscale texture and colour channels can be combined.

5. Memory calculation

For 2,448 × 2,048 pixels, total pixels = 5,013,504. An 8-bit grayscale array uses 5,013,504 bytes ≈

5.01 MB. A three-channel 8-bit RGB/BGR or HSV array uses 15,040,512 bytes ≈ 15.04 MB. Gray plus HSV held simultaneously uses approximately 20.05 MB before temporary arrays.

6. Recommended combination

Retain original colour for audit and colour variation. Convert to HSV for hue/saturation and value analysis. Generate grayscale for edges, texture and local contrast. Use a stable white balance and flat-field correction. Reuse buffers and process only the fabric region of interest to control memory and CPU load.

7. Task conclusion

No single representation is sufficient. The combined colour-plus-grayscale approach provides the best defect coverage while keeping the CPU pipeline manageable. The original image should be retained for traceability, while derived channels may be released after feature extraction.

Task-specific decision

The final choice is the approach that provides sufficient defect visibility and localisation while remaining explainable, maintainable and compatible with the available PC and budget. The recommendation must be verified on real fabric samples before production acceptance.

Evidence to collect during commissioning

- Capture defect-free fabric and each defect type under the selected lighting.
- Record known defect dimensions and confirm their pixel representation.
- Measure processing latency on the actual four-core PC.
- Validate width at multiple positions against a traceable reference.
- Compare detected defects with a human-labelled ground-truth map.
- Record false alarms, missed defects, marker position error and storage consumption.

TASK 3 – CALCULATIONS, IMPLEMENTATION AND OBSERVATIONS

Calculation / implementation notes

The numerical decision must be tied to the scenario: fabric speed, sampling, contrast, calibration, CPU time and marker position. No unmeasured accuracy or timing value is claimed.

Relevant OpenCV / NumPy implementation

```
# Generic CPU-side processing pattern
import cv2
import numpy as np
import time


def process_one(frame):
    start = time.perf_counter()
    blur = cv2.GaussianBlur(frame, (3, 3), 0.6)
    gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)
    local = cv2.GaussianBlur(gray, (31, 31), 0)
    residual = cv2.absdiff(gray, local)
    mask = (residual >= 5).astype(np.uint8)
    n, labels, stats, centres = cv2.connectedComponentsWithStats(
        mask, connectivity=8)
    elapsed_ms = (time.perf_counter() - start) * 1000
    return stats[1:], centres[1:], elapsed_ms


# Actual elapsed_ms must be measured on the target PC and camera image.
```

Observation and interpretation

The code demonstrates the structure of an image-processing experiment. It does not provide a factory detection score because no labelled fabric image dataset was supplied. During implementation, the output should be compared with known defect coupons and recorded in the validation tables in the appendices.

Limitations

- Thresholds are fabric-dependent and require recipe calibration.

- The sample code is a candidate generator, not a fully validated defect classifier.
- The actual camera frame rate, image format and transfer overhead must be measured.
- A defect decision must be converted to encoder position before the marker can act.

TASK 4 – Connectivity Analysis [4 Marks]

Aim and industrial objective

The aim of Task 4 is to compare the candidate approaches using the actual textile production constraints. The selected technique must support a 1,000 mm moving web, 1.5 mm minimum defect, 5–8 grey-level oil-stain contrast, ±2 mm width accuracy, a 0.5 s decision limit and a four-core CPU.

Key condition	Value	Why it matters
Web speed	0.6 m/s	Motion and timing
Minimum defect	1.5 mm	Sampling and optics
Width tolerance	±2 mm	Calibration
Compute	4-core CPU	Algorithm complexity
Budget	₹1,50,000	Component choice

1. 4-connectivity

A pixel is connected to its north, south, east and west neighbours. Diagonal pixels are not directly connected. This can split an irregular diagonal defect into multiple components, even when it appears visually continuous.

Approach	Strength	Limitation	Suitability
Candidate A	Simple / low cost	Limited context or performance	Low to moderate
Candidate B	Improves visibility or measurement	Needs calibration / extra processing	Moderate
Candidate C	More complete decision capability	Higher complexity or cost	Best when validated

2. 8-connectivity

A pixel is connected to all eight surrounding neighbours, including diagonals. It preserves diagonally connected defect boundaries and is useful for irregular holes, stains and thread fragments. It can merge separate defects that touch only at a corner.

3. m-connectivity

Mixed connectivity uses 4-neighbours and permits diagonal connections only when the common 4-neighbour pixels do not create an alternate foreground path. It reduces ambiguous diagonal links but is more complex to implement.

4. Worked binary pattern

Use the binary image: 0 1 0 0 0 / 1 0 0 1 0 / 0 0 1 0 0 / 0 0 0 1 1 / 0 0 0 0 0. Under 4-connectivity, diagonal-only pixels remain separate while the horizontal pair is one component. Under 8-connectivity, diagonal contacts merge. Under m-connectivity, diagonal merging depends on shared 4-neighbour occupancy.

5. Effect on measurements

Connectivity changes component count, area, bounding boxes and defect count. Fragmentation may create multiple tiny defects; merging may overestimate one defect's area. Area filters must therefore be combined with shape, distance and texture checks.

6. Recommendation and code

Use 8-connectivity for primary candidate labelling because rasterised textile defects often have diagonal boundaries. Apply shape and narrow-bridge checks to split corner-touching candidates when justified. OpenCV connectedComponentsWithStats supports connectivity=4 or 8; m-connectivity needs a custom neighbour traversal.

7. Task conclusion

8-connectivity is the practical default for irregular textile defects. m-connectivity is useful for topology-sensitive cases, while 4-connectivity is too likely to fragment diagonal defects.

Task-specific decision

The final choice is the approach that provides sufficient defect visibility and localisation while remaining explainable, maintainable and compatible with the available PC and budget. The recommendation must be verified on real fabric samples before production acceptance.

Evidence to collect during commissioning

- Capture defect-free fabric and each defect type under the selected lighting.
- Record known defect dimensions and confirm their pixel representation.

- Measure processing latency on the actual four-core PC.

- Validate width at multiple positions against a traceable reference.
- Compare detected defects with a human-labelled ground-truth map.
- Record false alarms, missed defects, marker position error and storage consumption.

TASK 4 – CALCULATIONS, IMPLEMENTATION AND OBSERVATIONS

Calculation / implementation notes

The numerical decision must be tied to the scenario: fabric speed, sampling, contrast, calibration, CPU time and marker position. No unmeasured accuracy or timing value is claimed.

Relevant OpenCV / NumPy implementation

```
# Generic CPU-side processing pattern
import cv2
import numpy as np
import time


def process_one(frame):
    start = time.perf_counter()
    blur = cv2.GaussianBlur(frame, (3, 3), 0.6)
    gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)
    local = cv2.GaussianBlur(gray, (31, 31), 0)
    residual = cv2.absdiff(gray, local)
    mask = (residual >= 5).astype(np.uint8)
    n, labels, stats, centres = cv2.connectedComponentsWithStats(
        mask, connectivity=8)
    elapsed_ms = (time.perf_counter() - start) * 1000
    return stats[1:], centres[1:], elapsed_ms


# Actual elapsed_ms must be measured on the target PC and camera image.
```

Observation and interpretation

The code demonstrates the structure of an image-processing experiment. It does not provide a factory detection score because no labelled fabric image dataset was supplied. During implementation, the output should be compared with known defect coupons and recorded in the validation tables in the appendices.

Limitations

- Thresholds are fabric-dependent and require recipe calibration.

- The sample code is a candidate generator, not a fully validated defect classifier.
- The actual camera frame rate, image format and transfer overhead must be measured.
- A defect decision must be converted to encoder position before the marker can act.

TASK 5 – Image Acquisition Analysis [5 Marks]

Aim and industrial objective

The aim of Task 5 is to compare the candidate approaches using the actual textile production constraints. The selected technique must support a 1,000 mm moving web, 1.5 mm minimum defect, 5–8 grey-level oil-stain contrast, ±2 mm width accuracy, a 0.5 s decision limit and a four-core CPU.

Key condition	Value	Why it matters
Web speed	0.6 m/s	Motion and timing
Minimum defect	1.5 mm	Sampling and optics
Width tolerance	±2 mm	Calibration
Compute	4-core CPU	Algorithm complexity
Budget	₹1,50,000	Component choice

1. Rolling shutter

A rolling-shutter sensor exposes or reads rows at different times. On a moving web, different rows correspond to different fabric positions, causing skew or geometric distortion. Short exposure reduces blur but does not remove row-to-row timing skew.

Approach	Strength	Limitation	Suitability
Candidate A	Simple / low cost	Limited context or performance	Low to moderate
Candidate B	Improves visibility or measurement	Needs calibration / extra processing	Moderate
Candidate C	More complete decision capability	Higher complexity or cost	Best when validated

2. Global shutter

A global-shutter sensor exposes all pixels at approximately the same instant. It freezes moving defects more consistently and is a good area-camera option. Exposure time and lighting still control motion blur.

3. Line scan

A line-scan camera captures one narrow line across the web. Successive lines build a 2D image as fabric moves. Encoder triggering keeps the machine-direction sampling consistent and makes line scan well suited to continuous web inspection.

4. Motion blur calculation

Blur distance $b = v \times t$. At 0.6 m/s, 1 ms exposure gives 0.6 mm blur; 0.5 ms gives 0.3 mm; 0.25 ms gives 0.15 mm. To keep blur below 10% of a 1.5 mm defect, $b \leq 0.15$ mm and $t \leq 0.15/600 = 0.00025$ s = 0.25 ms.

5. Comparison

Rolling shutter is often low-cost but can distort moving fabric. Global shutter gives strong frame geometry and is simpler to install than line scan. Line scan is best for continuous web inspection but needs an encoder, line-rate configuration and adequate bandwidth.

6. Line-rate estimate

At 4,096 pixels across 1,200 mm, horizontal sampling is 0.293 mm/px. For similar machine-direction sampling at 600 mm/s, line rate = $600/0.293 \approx 2{,}048$ lines/s. This is an indicative camera specification to verify against the interface and bit depth.

7. Task conclusion

Line scan is preferred for the continuous fabric web. A global-shutter area camera is the balanced fallback if line-scan procurement is too costly. Rolling shutter is not preferred for final quality inspection.

Task-specific decision

The final choice is the approach that provides sufficient defect visibility and localisation while remaining explainable, maintainable and compatible with the available PC and budget. The recommendation must be verified on real fabric samples before production acceptance.

Evidence to collect during commissioning

- Capture defect-free fabric and each defect type under the selected lighting.
- Record known defect dimensions and confirm their pixel representation.

- Measure processing latency on the actual four-core PC.
- Validate width at multiple positions against a traceable reference.

- Compare detected defects with a human-labelled ground-truth map.
- Record false alarms, missed defects, marker position error and storage consumption.

TASK 5 – CALCULATIONS, IMPLEMENTATION AND OBSERVATIONS

Calculation / implementation notes

Motion blur = speed × exposure. At 0.6 m/s and 0.25 ms, blur = 600 mm/s × 0.00025 s = 0.15 mm. This is 10% of a 1.5 mm defect.

Relevant OpenCV / NumPy implementation

```
# Generic CPU-side processing pattern
import cv2
import numpy as np
import time


def process_one(frame):
    start = time.perf_counter()
    blur = cv2.GaussianBlur(frame, (3, 3), 0.6)
    gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)
    local = cv2.GaussianBlur(gray, (31, 31), 0)
    residual = cv2.absdiff(gray, local)
    mask = (residual >= 5).astype(np.uint8)
    n, labels, stats, centres = cv2.connectedComponentsWithStats(
        mask, connectivity=8)
    elapsed_ms = (time.perf_counter() - start) * 1000
    return stats[1:], centres[1:], elapsed_ms


# Actual elapsed_ms must be measured on the target PC and camera image.
```

Observation and interpretation

The code demonstrates the structure of an image-processing experiment. It does not provide a factory detection score because no labelled fabric image dataset was supplied. During implementation, the output should be compared with known defect coupons and recorded in the validation tables in the appendices.

Limitations

- Thresholds are fabric-dependent and require recipe calibration.

- The sample code is a candidate generator, not a fully validated defect classifier.
- The actual camera frame rate, image format and transfer overhead must be measured.
- A defect decision must be converted to encoder position before the marker can act.

TASK 6 – Sampling Analysis [5 Marks]

Aim and industrial objective

The aim of Task 6 is to compare the candidate approaches using the actual textile production constraints. The selected technique must support a 1,000 mm moving web, 1.5 mm minimum defect, 5–8 grey-level oil-stain contrast, ±2 mm width accuracy, a 0.5 s decision limit and a four-core CPU.

Key condition	Value	Why it matters
Web speed	0.6 m/s	Motion and timing
Minimum defect	1.5 mm	Sampling and optics
Width tolerance	±2 mm	Calibration
Compute	4-core CPU	Algorithm complexity
Budget	₹1,50,000	Component choice

1. Pixel counts

1280×1024 = 1,310,720 pixels (1.31 MP). 2448×2048 = 5,013,504 pixels (5.01 MP). 4096×3000 = 12,288,000 pixels (12.29 MP).

Approach	Strength	Limitation	Suitability
Candidate A	Simple / low cost	Limited context or performance	Low to moderate
Candidate B	Improves visibility or measurement	Needs calibration / extra processing	Moderate
Candidate C	More complete decision capability	Higher complexity or cost	Best when validated

2. Spatial sampling

Assuming the horizontal image width maps to 1,200 mm, low resolution gives 1200/1280 = 0.9375 mm/px; medium gives 1200/2448 = 0.4902 mm/px; high gives 1200/4096 = 0.2930 mm/px.

3. Pixels across the 1.5 mm defect

Low: 1.5/0.9375 = 1.60 pixels. Medium: 1.5/0.4902 ≈ 3.06 pixels. High: 1.5/0.2930 ≈ 5.12 pixels. Low is inadequate; medium is borderline; high provides useful shape detail.

4. Fabric coverage

Fabric width/FOV width = 1000/1200 = 83.33%. If centred, nominal margin is 100 mm on each side. At 4,096 pixels across the FOV, fabric width occupies about 3,413 pixels.

5. Storage comparison

For 8-bit grayscale, image bytes equal pixel count: 1.31 MB, 5.01 MB and 12.29 MB. For 24-bit colour, multiply by three: 3.93 MB, 15.04 MB and 36.86 MB per image, uncompressed.

6. Decision

Select the 4,096-pixel horizontal option for the minimum defect because it gives about 5.12 pixels across 1.5 mm. Use ROI processing, compact masks and line-scan strip storage to reduce CPU and storage load. Medium resolution is a lower-cost fallback but has less sampling margin.

7. Task conclusion

High resolution best supports defect visibility. It must be balanced with data rate, CPU processing and seven-day retention. Resolution should be selected jointly with acquisition architecture and compression.

Task-specific decision

The final choice is the approach that provides sufficient defect visibility and localisation while remaining explainable, maintainable and compatible with the available PC and budget. The recommendation must be verified on real fabric samples before production acceptance.

Evidence to collect during commissioning

- Capture defect-free fabric and each defect type under the selected lighting.
- Record known defect dimensions and confirm their pixel representation.
- Measure processing latency on the actual four-core PC.
- Validate width at multiple positions against a traceable reference.
- Compare detected defects with a human-labelled ground-truth map.

- Record false alarms, missed defects, marker position error and storage consumption.

TASK 6 – CALCULATIONS, IMPLEMENTATION AND OBSERVATIONS

Calculation / implementation notes

mm/px = 1200 / horizontal pixels. Pixels per 1.5 mm defect = 1.5 / (mm/px). The resulting values are 1.60, 3.06 and 5.12 pixels for the three resolutions.

Relevant OpenCV / NumPy implementation

```
# Generic CPU-side processing pattern
import cv2
import numpy as np
import time


def process_one(frame):
    start = time.perf_counter()
    blur = cv2.GaussianBlur(frame, (3, 3), 0.6)
    gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)
    local = cv2.GaussianBlur(gray, (31, 31), 0)
    residual = cv2.absdiff(gray, local)
    mask = (residual >= 5).astype(np.uint8)
    n, labels, stats, centres = cv2.connectedComponentsWithStats(
        mask, connectivity=8)
    elapsed_ms = (time.perf_counter() - start) * 1000
    return stats[1:], centres[1:], elapsed_ms


# Actual elapsed_ms must be measured on the target PC and camera image.
```

Observation and interpretation

The code demonstrates the structure of an image-processing experiment. It does not provide a factory detection score because no labelled fabric image dataset was supplied. During implementation, the output should be compared with known defect coupons and recorded in the validation tables in the appendices.

Limitations

- Thresholds are fabric-dependent and require recipe calibration.

- The sample code is a candidate generator, not a fully validated defect classifier.
- The actual camera frame rate, image format and transfer overhead must be measured.
- A defect decision must be converted to encoder position before the marker can act.

TASK 7 – Quantization Analysis [4 Marks]

Aim and industrial objective

The aim of Task 7 is to compare the candidate approaches using the actual textile production constraints. The selected technique must support a 1,000 mm moving web, 1.5 mm minimum defect, 5–8 grey-level oil-stain contrast, ±2 mm width accuracy, a 0.5 s decision limit and a four-core CPU.

Key condition	Value	Why it matters
Web speed	0.6 m/s	Motion and timing
Minimum defect	1.5 mm	Sampling and optics
Width tolerance	±2 mm	Calibration
Compute	4-core CPU	Algorithm complexity
Budget	₹1,50,000	Component choice

1. Intensity levels

The number of quantization levels is 2^b. Four-bit provides 16 levels; six-bit provides 64; eight-bit provides 256; twelve-bit provides 4,096 levels.

Approach	Strength	Limitation	Suitability
Candidate A	Simple / low cost	Limited context or performance	Low to moderate
Candidate B	Improves visibility or measurement	Needs calibration / extra processing	Moderate
Candidate C	More complete decision capability	Higher complexity or cost	Best when validated

2. Oil-stain contrast

A 5–8 grey-level difference on an 8-bit scale is preserved nominally by 8-bit quantization. Six-bit quantization has a step of about 255/63 = 4.05 original levels, so a stain may become only one or two code steps. Four-bit quantization has a step of about 255/15 = 17 levels, so the stain contrast may disappear.

3. Storage and processing

Eight-bit data uses one byte per channel. Twelve-bit data needs 1.5 bytes per pixel when tightly packed, but many systems store it in 16-bit words. Reduced bit depth can save storage if packed, but common software arrays may expand values to 8-bit or 16-bit.

4. Sensor precision versus noise

More quantization levels do not guarantee better stain detection if sensor noise, exposure instability or lighting flicker is larger than the signal. The camera must have adequate signal-to-noise ratio and the illumination must be stable.

5. Recommendation

Acquire at 12-bit if affordable and supported, then map consistently to at least 8-bit for the CPU pipeline. Avoid per-frame automatic contrast stretching because it can make identical stains appear different in different frames.

6. OpenCV illustration

A quantization experiment should apply a fixed mapping to a test ramp containing 5–8 level differences, then compare preserved contrast at 4, 6, 8 and 12 bits. The resulting pixel values depend on the rounding convention and must be labelled as an illustrative software test.

7. Task conclusion

Eight-bit is the minimum practical processing depth. Twelve-bit acquisition provides useful headroom for correction; 4–6-bit reduction should not be used in the primary oil-stain detection path.

Task-specific decision

The final choice is the approach that provides sufficient defect visibility and localisation while remaining explainable, maintainable and compatible with the available PC and budget. The recommendation must be verified on real fabric samples before production acceptance.

Evidence to collect during commissioning

- Capture defect-free fabric and each defect type under the selected lighting.
- Record known defect dimensions and confirm their pixel representation.

- Measure processing latency on the actual four-core PC.
- Validate width at multiple positions against a traceable reference.

- Compare detected defects with a human-labelled ground-truth map.
- Record false alarms, missed defects, marker position error and storage consumption.

TASK 7 – CALCULATIONS, IMPLEMENTATION AND OBSERVATIONS

Calculation / implementation notes

Quantization levels = 2^bits. Four-bit = 16, six-bit = 64, eight-bit = 256, twelve-bit = 4096.

Relevant OpenCV / NumPy implementation

```
# Generic CPU-side processing pattern
import cv2
import numpy as np
import time


def process_one(frame):
    start = time.perf_counter()
    blur = cv2.GaussianBlur(frame, (3, 3), 0.6)
    gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)
    local = cv2.GaussianBlur(gray, (31, 31), 0)
    residual = cv2.absdiff(gray, local)
    mask = (residual >= 5).astype(np.uint8)
    n, labels, stats, centres = cv2.connectedComponentsWithStats(
        mask, connectivity=8)
    elapsed_ms = (time.perf_counter() - start) * 1000
    return stats[1:], centres[1:], elapsed_ms


# Actual elapsed_ms must be measured on the target PC and camera image.
```

Observation and interpretation

The code demonstrates the structure of an image-processing experiment. It does not provide a factory detection score because no labelled fabric image dataset was supplied. During implementation, the output should be compared with known defect coupons and recorded in the validation tables in the appendices.

Limitations

- Thresholds are fabric-dependent and require recipe calibration.
- The sample code is a candidate generator, not a fully validated defect classifier.

- The actual camera frame rate, image format and transfer overhead must be measured.
- A defect decision must be converted to encoder position before the marker can act.

TASK 8 – Image Formation Analysis [5 Marks]

Aim and industrial objective

The aim of Task 8 is to compare the candidate approaches using the actual textile production constraints. The selected technique must support a 1,000 mm moving web, 1.5 mm minimum defect, 5–8 grey-level oil-stain contrast, ±2 mm width accuracy, a 0.5 s decision limit and a four-core CPU.

Key condition	Value	Why it matters
Web speed	0.6 m/s	Motion and timing
Minimum defect	1.5 mm	Sampling and optics
Width tolerance	±2 mm	Calibration
Compute	4-core CPU	Algorithm complexity
Budget	₹1,50,000	Component choice

1. Image formation at 700 mm

Light from the LED source reflects from the fabric and passes through the lens onto the sensor. The pinhole-camera model maps 3D points to image coordinates. Because the fabric is approximately planar, a homography can map pixels to millimetres after calibration.

Approach	Strength	Limitation	Suitability
Candidate A	Simple / low cost	Limited context or performance	Low to moderate
Candidate B	Improves visibility or measurement	Needs calibration / extra processing	Moderate
Candidate C	More complete decision capability	Higher complexity or cost	Best when validated

2. Perspective and lens distortion

Camera tilt causes scale variation across the field. Lens distortion can curve straight fabric edges. Camera height alone does not determine pixel scale; focal length, sensor size, focus and mounting

angle are also needed. These values were not provided, so an exact focal-length calculation is not possible.

3. Calibration procedure

Place a known checkerboard or dot-grid target at the fabric plane. Capture it at multiple field positions. Estimate camera intrinsics and distortion. Fit a planar homography. Validate using an independent certified 1,000 mm reference and gauge lengths.

4. Width accuracy

The required ±2 mm error on 1,000 mm is 2/1000×100 = 0.2%. Fit the left and right fabric edges over multiple rows and convert them through the calibrated mapping. Repeat at left, centre and right field positions.

5. Lighting range

The lux range changes by 3000/500 = 6 times. A fixed global threshold may fail as illumination changes. High light can saturate yarns and reduce stain contrast; low light increases noise and hides broken threads.

6. Recommended arrangement

Use a rigid normal camera mount, low-distortion lens, flat fabric support, diffuse flicker-free LED enclosure and flat-field reference. Add low-angle lighting if raised or broken yarns are difficult to see. Monitor reference intensity to detect lighting drift.

7. Task conclusion

Calibrated imaging and controlled illumination are necessary for ±2 mm width accuracy and low-contrast stain detection. Calibration corrects optical geometry but cannot correct physical fabric stretching.

Task-specific decision

The final choice is the approach that provides sufficient defect visibility and localisation while remaining explainable, maintainable and compatible with the available PC and budget. The recommendation must be verified on real fabric samples before production acceptance.

Evidence to collect during commissioning

- Capture defect-free fabric and each defect type under the selected lighting.

- Record known defect dimensions and confirm their pixel representation.

- Measure processing latency on the actual four-core PC.
- Validate width at multiple positions against a traceable reference.
- Compare detected defects with a human-labelled ground-truth map.
- Record false alarms, missed defects, marker position error and storage consumption.

TASK 8 – CALCULATIONS, IMPLEMENTATION AND OBSERVATIONS

Calculation / implementation notes

Relative width tolerance = 2 mm / 1000 mm × 100 = 0.2%. The calibration must be validated across the full image field.

Relevant OpenCV / NumPy implementation

```
# Generic CPU-side processing pattern
import cv2
import numpy as np
import time


def process_one(frame):
    start = time.perf_counter()
    blur = cv2.GaussianBlur(frame, (3, 3), 0.6)
    gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)
    local = cv2.GaussianBlur(gray, (31, 31), 0)
    residual = cv2.absdiff(gray, local)
    mask = (residual >= 5).astype(np.uint8)
    n, labels, stats, centres = cv2.connectedComponentsWithStats(
        mask, connectivity=8)
    elapsed_ms = (time.perf_counter() - start) * 1000
    return stats[1:], centres[1:], elapsed_ms


# Actual elapsed_ms must be measured on the target PC and camera image.
```

Observation and interpretation

The code demonstrates the structure of an image-processing experiment. It does not provide a factory detection score because no labelled fabric image dataset was supplied. During implementation, the output should be compared with known defect coupons and recorded in the validation tables in the appendices.

Limitations

- Thresholds are fabric-dependent and require recipe calibration.

- The sample code is a candidate generator, not a fully validated defect classifier.
- The actual camera frame rate, image format and transfer overhead must be measured.
- A defect decision must be converted to encoder position before the marker can act.

TASK 9 – Implementation and Performance Evaluation [6 Marks]

Aim and industrial objective

The aim of Task 9 is to compare the candidate approaches using the actual textile production constraints. The selected technique must support a 1,000 mm moving web, 1.5 mm minimum defect, 5–8 grey-level oil-stain contrast, ±2 mm width accuracy, a 0.5 s decision limit and a four-core CPU.

Key condition	Value	Why it matters
Web speed	0.6 m/s	Motion and timing
Minimum defect	1.5 mm	Sampling and optics
Width tolerance	±2 mm	Calibration
Compute	4-core CPU	Algorithm complexity
Budget	₹1,50,000	Component choice

1. System architecture

Diffuse LED and optional low-angle light illuminate the fabric. Camera and encoder acquire images. The four-core PC performs preprocessing, segmentation, classification and measurement. Defect records are stored locally and transmitted over Gigabit Ethernet to a PLC, which schedules the marking mechanism.

Approach	Strength	Limitation	Suitability
Candidate A	Simple / low cost	Limited context or performance	Low to moderate
Candidate B	Improves visibility or measurement	Needs calibration / extra processing	Moderate
Candidate C	More complete decision capability	Higher complexity or cost	Best when validated

2. Three configurations

Budget: minimum-cost rolling-shutter camera, basic lighting and simple CPU thresholds. Balanced: global-shutter or economical line-scan camera, controlled LED illumination, encoder and optimised

three-level CPU pipeline. Premium: high-resolution industrial line-scan, advanced lighting and higher-performance compute.

3. Indicative cost comparison

Budget camera/lens/light package may be planned at ₹35,000–₹55,000. Balanced configuration may be planned at ₹75,000–₹1,05,000 for camera, optics and lighting, with full integration allocation around ₹1.40 lakh. Premium industrial camera and advanced compute may exceed ₹1.50 lakh. These are planning estimates, not quotations.

4. Defect recording

Holes use local contrast, edge closure and region shape. Broken threads use directional continuity. Oil stains use local brightness and saturation deviations. Weaving irregularities use yarn spacing and texture periodicity. Colour variations use calibrated colour differences. Each record includes class, confidence, bounding box, centroid and encoder position.

5. Marker communication

At 0.6 m/s the 0.3 m travel time is 0.5 s. If the encoder position at capture is E, the marker target is E+300 mm. The PLC activates the marker when the encoder reaches that position, compensating for actuator delay. A fixed timer is less reliable when speed varies.

6. Storage calculation

A 4096×3000 8-bit grayscale frame is 12.288 MB. At 2 frames/s for seven days, raw storage is 12.288×2×604800 = 14.86 TB. For a 4096-pixel line at 2048 lines/s, raw grayscale is about 8.39 MB/s, or 5.07 TB per week. A verified 4:1 compression gives about 1.27 TB, leaving some margin within 2 TB.

7. Task conclusion

The balanced configuration is the most practical within ₹1.50 lakh, assuming the PC and storage are already available. It needs measured compression and latency validation. If the requirement means all raw uncompressed frames, 2 TB is insufficient.

Task-specific decision

The final choice is the approach that provides sufficient defect visibility and localisation while remaining explainable, maintainable and compatible with the available PC and budget. The

recommendation must be verified on real fabric samples before production acceptance.

Evidence to collect during commissioning

- Capture defect-free fabric and each defect type under the selected lighting.
- Record known defect dimensions and confirm their pixel representation.
- Measure processing latency on the actual four-core PC.
- Validate width at multiple positions against a traceable reference.
- Compare detected defects with a human-labelled ground-truth map.
- Record false alarms, missed defects, marker position error and storage consumption.

TASK 9 – CALCULATIONS, IMPLEMENTATION AND OBSERVATIONS

Calculation / implementation notes

Marker travel time = 0.3/0.6 = 0.5 s. This leaves no allowance beyond the stated decision limit, so the internal processing target must be earlier than 0.5 s.

Relevant OpenCV / NumPy implementation

```
# Generic CPU-side processing pattern
import cv2
import numpy as np
import time


def process_one(frame):
    start = time.perf_counter()
    blur = cv2.GaussianBlur(frame, (3, 3), 0.6)
    gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)
    local = cv2.GaussianBlur(gray, (31, 31), 0)
    residual = cv2.absdiff(gray, local)
    mask = (residual >= 5).astype(np.uint8)
    n, labels, stats, centres = cv2.connectedComponentsWithStats(
        mask, connectivity=8)
    elapsed_ms = (time.perf_counter() - start) * 1000
    return stats[1:], centres[1:], elapsed_ms


# Actual elapsed_ms must be measured on the target PC and camera image.
```

Observation and interpretation

The code demonstrates the structure of an image-processing experiment. It does not provide a factory detection score because no labelled fabric image dataset was supplied. During implementation, the output should be compared with known defect coupons and recorded in the validation tables in the appendices.

Limitations

- Thresholds are fabric-dependent and require recipe calibration.

- The sample code is a candidate generator, not a fully validated defect classifier.
- The actual camera frame rate, image format and transfer overhead must be measured.
- A defect decision must be converted to encoder position before the marker can act.

TASK 10 – Front-End Adaptation Challenge [8 Marks]

Aim and industrial objective

The aim of Task 10 is to compare the candidate approaches using the actual textile production constraints. The selected technique must support a 1,000 mm moving web, 1.5 mm minimum defect, 5–8 grey-level oil-stain contrast, ±2 mm width accuracy, a 0.5 s decision limit and a four-core CPU.

Key condition	Value	Why it matters
Web speed	0.6 m/s	Motion and timing
Minimum defect	1.5 mm	Sampling and optics
Width tolerance	±2 mm	Calibration
Compute	4-core CPU	Algorithm complexity
Budget	₹1,50,000	Component choice

1. Changed requirements

The new order requires 0.4 mm defects, UV-visible stains, 1.2 m/s speed, twice current volume and 0.3 s decision time. These changes affect spatial sampling, camera response, exposure, line rate, compute load, storage and marker timing.

Approach	Strength	Limitation	Suitability
Candidate A	Simple / low cost	Limited context or performance	Low to moderate
Candidate B	Improves visibility or measurement	Needs calibration / extra processing	Moderate
Candidate C	More complete decision capability	Higher complexity or cost	Best when validated

2. Inadequate existing components

The 4,096-pixel horizontal option gives 0.293 mm/px, so a 0.4 mm defect spans only 1.37 pixels. Visible-only lighting and a non-UV-sensitive sensor cannot reveal UV-only stains. The existing PC

may not process twice the data in 0.3 s.

3. New sampling requirement

For four pixels across 0.4 mm, pixel pitch must be ≤0.1 mm. Across 1,200 mm, horizontal sampling must be at least 12,000 pixels. For five pixels across, pitch is 0.08 mm and width sampling is 15,000 pixels.

4. Speed and exposure

At 1.2 m/s and 0.1 mm machine-direction pitch, line rate = 1200/0.1 = 12,000 lines/s. To limit blur to 10% of a 0.4 mm defect, blur ≤0.04 mm and exposure ≤0.04/1200 = 33 microseconds.

5. UV channel

UV illumination may excite or reveal stains not visible under white light. The camera must have UV response and suitable optics/filters; a conventional colour camera may block UV. A dedicated UV-sensitive sensor or additional channel, emission filtering, shielding and interlocks may be required.

6. Strategy comparison

Retain/software-only is not feasible because software cannot create missing spatial samples or UV sensitivity. Modify selected components is a practical short-term compromise if a suitable camera and UV channel fit the budget, but the PC may remain a bottleneck. Redesign with high-resolution line scan, camera-side preprocessing and low-latency marker architecture is technically strongest but needs additional resources.

7. Task conclusion

Under unchanged PC and budget, selected-component modification is the practical risk-reduction strategy, not a guarantee of full compliance. Full compliance likely requires a redesign, more compute/storage, a closer marker or a mechanical buffer.

Task-specific decision

The final choice is the approach that provides sufficient defect visibility and localisation while remaining explainable, maintainable and compatible with the available PC and budget. The recommendation must be verified on real fabric samples before production acceptance.

Evidence to collect during commissioning

- Capture defect-free fabric and each defect type under the selected lighting.

- Record known defect dimensions and confirm their pixel representation.

- Measure processing latency on the actual four-core PC.
- Validate width at multiple positions against a traceable reference.
- Compare detected defects with a human-labelled ground-truth map.
- Record false alarms, missed defects, marker position error and storage consumption.

TASK 10 – CALCULATIONS, IMPLEMENTATION AND OBSERVATIONS

Calculation / implementation notes

New marker travel time = 0.3/1.2 = 0.25 s. A 0.3 s decision would be late by 0.05 s. The physical layout or decision deadline must change.

Relevant OpenCV / NumPy implementation

```
# Generic CPU-side processing pattern
import cv2
import numpy as np
import time


def process_one(frame):
    start = time.perf_counter()
    blur = cv2.GaussianBlur(frame, (3, 3), 0.6)
    gray = cv2.cvtColor(blur, cv2.COLOR_BGR2GRAY)
    local = cv2.GaussianBlur(gray, (31, 31), 0)
    residual = cv2.absdiff(gray, local)
    mask = (residual >= 5).astype(np.uint8)
    n, labels, stats, centres = cv2.connectedComponentsWithStats(
        mask, connectivity=8)
    elapsed_ms = (time.perf_counter() - start) * 1000
    return stats[1:], centres[1:], elapsed_ms


# Actual elapsed_ms must be measured on the target PC and camera image.
```

Observation and interpretation

The code demonstrates the structure of an image-processing experiment. It does not provide a factory detection score because no labelled fabric image dataset was supplied. During implementation, the output should be compared with known defect coupons and recorded in the validation tables in the appendices.

Limitations

- Thresholds are fabric-dependent and require recipe calibration.

- The sample code is a candidate generator, not a fully validated defect classifier.
- The actual camera frame rate, image format and transfer overhead must be measured.
- A defect decision must be converted to encoder position before the marker can act.

APPENDIX A – DEFECT TAXONOMY AND ANNOTATION

Supplementary engineering detail for the Sri Lakshmi Textile Industries inspection project.

Holes

Missing fabric material may reveal the background or conveyor. Use contrast, contour closure and known-size validation. Avoid confusing open weave or lint gaps with true holes.

Broken threads

A warp or weft yarn is interrupted. Use directional filtering and continuity analysis. Compare the expected thread direction with local line response.

Oil stains

Oil may darken the fabric, alter saturation or change reflectance. Use flat-field correction and combined intensity/colour evidence.

Weaving irregularities

Skipped picks, floats, density changes or uneven spacing alter periodic texture. Compare local yarn spacing and orientation against the approved fabric recipe.

Colour variation

A patch differs from approved shade or surrounding fabric. Use calibrated colour references and compare local colour statistics.

Practical record

The commissioning team should record the actual measurement, date, instrument and responsible person. The report intentionally leaves factory measurements blank until they can be collected from the real line.

APPENDIX B – CALIBRATION AND WIDTH VALIDATION

Supplementary engineering detail for the Sri Lakshmi Textile Industries inspection project.

Calibration sequence

Warm up camera and LEDs; clean optics; place a certified grid at fabric height; capture across the field; estimate intrinsic and distortion parameters; fit a planar homography; validate with an independent gauge.

Width calculation

Measured width W = x_right − x_left after converting both edge coordinates into millimetres. Error = measured width − reference width. Pass if absolute error ≤2 mm.

Position testing

Measure a known-width target at left, centre and right positions. A single centre measurement cannot reveal field-dependent lens or perspective error.

Physical fabric variation

Calibration corrects imaging geometry but not real stretching, skew or tension changes. A defined tension and support condition is required.

Practical record

The commissioning team should record the actual measurement, date, instrument and responsible person. The report intentionally leaves factory measurements blank until they can be collected from the real line.

APPENDIX C – LIGHTING AND STAIN VISIBILITY

Supplementary engineering detail for the Sri Lakshmi Textile Industries inspection project.

Illuminance range

3000 lux / 500 lux = 6. The lighting variation can be larger than the subtle stain contrast and can make global thresholds unreliable.

Diffuse lighting

A diffuse LED panel or bar gives uniform illumination and reduces harsh shadows. Avoid clipping bright yarn surfaces.

Dark-field lighting

Low-angle light can emphasise raised threads and surface defects, but may also highlight harmless lint and weave texture.

Polarisation

Cross-polarised illumination may reduce glare from reflective fibres or finishes, but reduces available light and requires testing.

Acceptance test

Use a stable low-contrast target with a 5–8 level difference and verify that the target remains detectable across the field and operating time.

Practical record

The commissioning team should record the actual measurement, date, instrument and responsible person. The report intentionally leaves factory measurements blank until they can be collected from the real line.

APPENDIX D – ALGORITHM AND FEATURE DESIGN

Supplementary engineering detail for the Sri Lakshmi Textile Industries inspection project.

Local contrast

Compare each pixel with a nearby estimated background instead of a single global threshold.

Texture

Use local variance, directional gradients, line continuity and thread spacing to distinguish weave irregularity from stains.

Shape

Use area, aspect ratio, perimeter, compactness and orientation to distinguish hole-like and thread-like regions.

Colour

Use HSV and calibrated colour differences; down-weight hue where saturation is very low.

Unknown anomalies

Low-confidence or unfamiliar regions should be saved and reviewed rather than automatically accepted.

Practical record

The commissioning team should record the actual measurement, date, instrument and responsible person. The report intentionally leaves factory measurements blank until they can be collected from the real line.

APPENDIX E – VALIDATION DATASET AND METRICS

Supplementary engineering detail for the Sri Lakshmi Textile Industries inspection project.

Dataset composition

Include normal fabric, holes, broken threads, oil stains, weaving irregularities, colour variations and hard negatives such as seams, lint, folds and shadows.

Split policy

Split by roll or production batch, not random adjacent frames, to prevent near-duplicate leakage between training and test sets.

Miss rate

Missed-defect rate = FN/(TP+FN) × 100. Required maximum is 2%.

Precision and recall

Precision = TP/(TP+FP); recall = TP/(TP+FN). Report per defect class as well as overall.

Width error

Compare camera width measurement against a traceable reference at multiple locations and repeat runs.

Practical record

The commissioning team should record the actual measurement, date, instrument and responsible person. The report intentionally leaves factory measurements blank until they can be collected from the real line.

APPENDIX F – STORAGE AND NETWORK

Supplementary engineering detail for the Sri Lakshmi Textile Industries inspection project.

Area-frame storage

A 4096×3000 8-bit grayscale frame is 12.288 MB. At 2 fps for seven days this is approximately 14.86 TB uncompressed.

Line-scan storage

4096 pixels × 2048 lines/s × 1 byte ≈ 8.39 MB/s; seven days is approximately 5.07 TB raw. A verified 4:1 compression gives about 1.27 TB.

Retention controls

Monitor disk space, reserve capacity for metadata and defect crops, and do not delete images before seven days.

PLC message

Transmit defect ID, class, confidence, x coordinate, encoder target and timestamp. Use sequence numbers and acknowledgements to avoid duplicate or lost marks.

Practical record

The commissioning team should record the actual measurement, date, instrument and responsible person. The report intentionally leaves factory measurements blank until they can be collected from the real line.

APPENDIX G – SAFETY AND FAILURE HANDLING

Supplementary engineering detail for the Sri Lakshmi Textile Industries inspection project.

Camera failure

Raise alarm and stop automatic pass decisions; switch to manual inspection or safe stop.

Lighting failure

Detect reference intensity drift and inhibit automatic decisions if the image is not valid.

Encoder failure

Disable position-based marking and request safe handling.

Storage full

Alert before the retention window is threatened; never silently overwrite required audit evidence.

PC overload

Monitor queue depth and latency; reduce noncritical operations and alarm when the timing limit is exceeded.

UV safety for new order

Enclose UV sources, use interlocks and verify material compatibility and operator exposure controls.

Practical record

The commissioning team should record the actual measurement, date, instrument and responsible person. The report intentionally leaves factory measurements blank until they can be collected from the real line.

APPENDIX H – DASHBOARD AND MAINTENANCE

Supplementary engineering detail for the Sri Lakshmi Textile Industries inspection project.

Dashboard fields

Show roll ID, recipe, current width, defect counts by class, defect list, confidence, mark status, average and maximum latency, storage remaining and equipment health.

Daily maintenance

Clean lens/window, inspect LEDs, run a reference sample, check encoder pulses, verify PLC communication and review unknown detections.

Version control

Record software version, threshold set, camera calibration ID and operator action for every decision.

Periodic audit

Inspect a sample of accepted fabric and compare with automated decisions to detect drift or new defect types.

Practical record

The commissioning team should record the actual measurement, date, instrument and responsible person. The report intentionally leaves factory measurements blank until they can be collected from the real line.

APPENDIX I – FORMULA SUMMARY

Supplementary engineering detail for the Sri Lakshmi Textile Industries inspection project.

Spatial sampling

mm/pixel = FOV width / horizontal pixel count.

Defect pixels

pixels across defect = defect width / mm per pixel.

Motion blur

blur distance = conveyor speed × exposure time.

Line rate

lines per second = web speed / machine-direction pixel pitch.

Quantization

intensity levels = 2^number_of_bits.

Width tolerance

relative error = allowed error / nominal width × 100%.

Marker time

travel time = camera-to-marker distance / web speed.

Practical record

The commissioning team should record the actual measurement, date, instrument and responsible person. The report intentionally leaves factory measurements blank until they can be collected from the real line.

APPENDIX J – COMMISSIONING CHECKLIST AND CONCLUSION

Supplementary engineering detail for the Sri Lakshmi Textile Industries inspection project.

Bench test

Validate focus, field of view, calibration, illumination and minimum defect pixel representation.

Slow-speed trial

Run representative rolls at reduced speed and compare output with manual ground truth.

Rated-speed trial

Run at 0.6 m/s; measure latency, dropped frames, queue depth, encoder position and marker error.

Acceptance

Demonstrate missed-defect rate ≤2%, width error ≤±2 mm, decision time ≤0.5 s and seven-day retention.

Overall conclusion

The balanced calibrated computer-vision configuration is the most practical initial design. The new export requirements cannot all be guaranteed with the unchanged PC, budget and marker position.

Practical record

The commissioning team should record the actual measurement, date, instrument and responsible person. The report intentionally leaves factory measurements blank until they can be collected from the real line.

COMMISSIONING RECORD – FACTORY MEASUREMENTS TO BE COMPLETED

Record item	Measured value / notes	Date / initials
Camera make/model and serial number		
Lens and focal length		
Mounting height		
Measured field of view		
Illuminance at left edge		
Illuminance at centre		
Illuminance at right edge		
Exposure time		
Encoder pulses per metre		
Horizontal mm/pixel		
Vertical mm/pixel		
Width error at left		
Width error at centre		
Width error at right		
Mean decision latency		
Maximum decision latency		
Compression ratio		
Seven-day storage estimate		
Missed-defect rate		
Marker position error		

Vision engineer: _____ Production supervisor: _____

Quality manager: _____ Date: _____

VIVA PREPARATION – TASK-WISE QUESTIONS

Why is automated inspection preferred? It supports repeatable continuous inspection, measurement, traceability and automatic marking.

Why use a three-level pipeline? It combines pixel correction, meaningful features and a final decision while remaining explainable.

Why not grayscale alone? It loses chromatic information needed for colour variation.

Why use 8-connectivity? It preserves diagonal continuity in irregular defect regions.

Why is line scan suitable? It is designed for continuous moving webs and can synchronise to an encoder.

Why is 1280-pixel sampling inadequate? A 1.5 mm defect spans only 1.6 pixels.

Why retain 8-bit or 12-bit data? A 5–8 level stain contrast can be lost by coarse quantization.

Why calibrate? Camera height alone does not define a correct pixel-to-mm scale.

Why use encoder-based marking? It follows actual fabric motion even when speed varies.

Can software alone meet the new order? No; it cannot create missing spatial samples, UV sensor response or additional compute.

DECLARATION, REFERENCES AND LIMITATIONS

This report is based on the industrial scenario and numerical specifications supplied for Computer Vision Assignment 2. Additional assumptions are stated where necessary. Numerical results are calculations or engineering estimates unless explicitly identified as actual measurements.

The code snippets are implementation templates and require testing on real fabric images. Example event records and dashboard rows are illustrative. Cost allocations are indicative planning estimates, not vendor quotations. No factory detection accuracy, measured CPU latency or physical installation result is claimed.

Production acceptance requires real fabric samples, calibrated dimensional targets, actual camera timing, a representative defect dataset and a controlled marker trial. The 2% missed-defect limit and ±2 mm width accuracy must be demonstrated experimentally.

References

- R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th Edition, Pearson.
- R. Szeliski, Computer Vision: Algorithms and Applications, 2nd Edition, Springer.
- OpenCV Documentation, image processing and camera calibration modules, https://docs.opencv.org/.
- EMVA 1288, Standard for Characterization of Image Sensors and Cameras.
- ISO 25178 series, Geometrical product specifications – Areal surface texture.

Student signature: _____Date:

_____

TECHNICAL EXPANSION 01 – Inspection strategy and production-quality requirements

Purpose. Translate the textile mill's production objective into measurable inspection-system requirements.

Engineering analysis

Manual inspection depends on visual attention, experience and fatigue. It can recognise unfamiliar defects but is difficult to standardise over long shifts and cannot consistently output encoder-referenced coordinates. Assisted inspection improves repeatability by presenting highlighted regions to an operator, but human confirmation introduces a variable delay. Fully automated inspection applies the same acquisition and decision rules to every inspected region and can issue a marker command without waiting for a person. For a continuously moving web at 0.6 m/s, the automated approach is the only candidate that naturally supports continuous detection and automatic marking. A practical installation should nevertheless retain an operator review queue for uncertain regions and periodic audits.

Worked calculation / design note

At 0.6 m/s, fabric travels 600 mm each second. The camera-to-marker distance is 300 mm, so the nominal travel time is 300/600 = 0.5 s. The stated maximum decision time is also 0.5 s. This leaves effectively no allowance for actuator response, communication jitter or position uncertainty if the decision deadline is interpreted as the time available before the defect reaches the marker. The control design must therefore use encoder position prediction rather than relying on a fixed delay alone. This calculation is a key commissioning risk.

Verification procedure

Conduct a shift-length trial with seeded defects and compare automatic results with an independent reference inspection. Record class-wise recall, false alarms per metre, operator review time, marker success and width error. Include both cotton and cotton-polyester recipes.

Evidence to include in the final submission

Prepare a requirements traceability table linking every quality requirement to a sensor, algorithm, output and acceptance test.

Parameter / evidence	Value to record	Remarks / status

Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 02 – Human inspection: strengths, weaknesses and audit role

Purpose. Assess where trained operators remain valuable within an automated textile inspection cell.

Engineering analysis

Human inspectors use direct visual observation under suitable lighting. They can use context, experience and tactile examination to distinguish a true hole from a normal weave opening. Their strengths include adaptability and the ability to investigate unusual events. Their limitations include fatigue, inconsistent judgement, restricted reaction time and difficulty recording exact defect coordinates. Human inspection is therefore valuable as an audit and escalation layer, not as the primary real-time marker controller. The system should show the original image, defect crop, proposed class, confidence and machine position so that an operator can make a traceable decision.

Worked calculation / design note

Suppose an operator needs 2 seconds to inspect and record a candidate region. During this time the web travels $0.6 \times 2 = 1.2$ m. This is four times the 0.3 m camera-to-marker separation. Thus a manual decision cannot reliably be made before the same defect reaches the marker unless the fabric is stopped or buffered. This is an illustrative timing calculation, not a measured operator speed.

Verification procedure

Run a blind review exercise with two inspectors using the same sample set. Compare their class labels and defect locations. Record disagreements and use them to improve the defect taxonomy and training examples.

Evidence to include in the final submission

Create an operator review protocol defining accept, reject, uncertain, reclassify and escalate actions.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	

Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 03 – Assisted inspection and human-in-the-loop control

Purpose. Design an assisted workflow that adds human oversight without blocking the real-time inspection path.

Engineering analysis

An assisted system processes images automatically and displays suspected defects to an operator. It can be useful during commissioning, when thresholds are being tuned, or when the mill introduces a new fabric recipe. However, the operator should not be required to approve every defect before the PLC receives a time-critical marker command. A safer design separates the real-time path from the review path: the machine makes a conservative decision using validated rules, while low-confidence examples are placed in a queue for later review. The original automatic result must remain stored even if the operator changes the label.

Worked calculation / design note

Use three decision bands as a starting design assumption: high confidence is automatically marked; medium confidence is marked and queued for review; low confidence triggers a line alert or a defined fail-safe action. Numeric confidence thresholds must be calibrated using labelled production data and should not be invented before validation. The system should record threshold version and operator override reason.

Verification procedure

Test the queue under normal production and under a deliberately difficult fabric batch. Check that queue growth does not slow the real-time inspection thread. Confirm that operator actions do not rewrite the original event history.

Evidence to include in the final submission

Document the review screen fields, escalation rules and audit log format.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	

Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 04 – Automated inspection: reliability and failure containment

Purpose. Explain how automation can deliver repeatable inspection while controlling the risks of missed defects.

Engineering analysis

Automation provides consistent acquisition and deterministic processing when camera, exposure, lighting and recipe settings are controlled. It can calculate fabric width, detect defect candidates and report positions. Its principal risks are domain shift, incorrect calibration, sensor failure, lighting drift and software faults. It may miss a defect not represented in its rules or training data. Reliability therefore requires more than a classifier: it requires reference checks, watchdogs, encoder validation, storage monitoring, confidence handling and a clear stop-or-alert policy.

Worked calculation / design note

The missed-defect requirement is at most 2%. If a validation set contains 500 true defects, the maximum permitted number of misses is 10 because $10/500 \times 100 = 2\%$. A more convincing acceptance test should include enough examples of each defect class; a large total count can conceal poor performance on a rare class. For example, 2 misses among 20 oil stains is a 10% miss rate for that class even if overall recall is high.

Verification procedure

Use a hold-out test set that was not used to tune thresholds. Report class-wise recall and a confidence interval. Re-test after camera relocation, lighting changes or a major software update.

Evidence to include in the final submission

Prepare a failure-mode register and an acceptance report with per-class missed-defect rates.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 05 – Inspection strategy decision matrix and commissioning plan

Purpose. Consolidate the three approaches into a defensible selection for the stated textile line.

Engineering analysis

Manual inspection is flexible but labour intensive and weak for precise real-time marking. Assisted inspection improves visibility and consistency but retains a human bottleneck. Fully automated vision best fits continuous production, provided that uncertain cases are escalated and the line has a fail-safe response. The selection should be based on measured performance rather than claims that automation is perfect. The production manager should approve the risk controls before the system is connected to the marker.

Worked calculation / design note

Use a weighted decision matrix with criteria such as minimum defect detection, repeatability, response time, width measurement, labour demand, traceability and capital cost. Assign weights with the mill's quality team. Score each approach on the same scale and preserve the reasoning behind each score. The final recommendation is expected to favour automation because the marker needs a machine-readable defect position within the specified time.

Verification procedure

Conduct a pilot on representative cotton and cotton-polyester rolls. Include normal fabric, known defects, edge fraying, colour changes and lighting disturbances. Compare system events with a reference inspection map.

Evidence to include in the final submission

Submit the weighted matrix, risk register, pilot results and final sign-off.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 06 – Low-level image processing pipeline design

Purpose. Specify the acquisition, preprocessing and enhancement stages needed before defect analysis.

Engineering analysis

The low-level pipeline converts sensor output into a stable image suitable for later analysis. Acquisition obtains the frame or line data. Preprocessing may include black-level correction, flat-field correction, mild denoising and colour conversion. Enhancement can improve local contrast, but aggressive enhancement can create false edges or erase small defects. A low-level pipeline alone does not make a final defect decision because it does not explicitly identify regions, measure their properties or assign defect classes. It is best viewed as the foundation of a complete pipeline.

Worked calculation / design note

At the stated 1,200 mm field width, a 2448-pixel-wide image samples approximately 1200/2448 = 0.4902 mm per pixel. A 1.5 mm defect spans about 1.5/0.4902 = 3.06 pixels horizontally. This is only a sampling estimate; optical blur and defect orientation can reduce useful visibility. The low-level stage must preserve these few pixels and avoid large smoothing kernels.

Verification procedure

Capture a known defect coupon before and after each preprocessing operation. Compare local contrast and boundary visibility. Reject any filter that causes a known 1.5 mm defect to disappear or merge with the background.

Evidence to include in the final submission

Provide a pipeline diagram, parameter table and before/after sample images.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 07 – Intermediate feature extraction for textile defects

Purpose. Describe features that distinguish holes, broken threads, oil stains and weave irregularities.

Engineering analysis

Intermediate processing converts pixels into meaningful candidate regions and measurements. Useful features include local contrast, area, perimeter, bounding-box dimensions, aspect ratio, orientation, edge strength, colour statistics and texture energy. A hole may form a compact or irregular region with a strong boundary. A broken thread often forms a narrow elongated discontinuity aligned with warp or weft direction. Oil stains may have irregular boundaries and local intensity or saturation differences. Weaving irregularities are often better represented by texture and periodicity features than by simple colour thresholds.

Worked calculation / design note

Suppose a candidate occupies 18 pixels and the calibrated pixel area is $0.4902 \times 0.1333 = 0.06535$ mm² for an example 2448×3000 image over 1200×400 mm. Its estimated area is $18 \times 0.06535 = 1.1763$ mm². This area is only a feature estimate; the region's physical shape and threshold sensitivity must also be considered.

Verification procedure

Create annotated regions for every defect class and calculate feature distributions. Check whether one feature separates classes or whether a combination is required. Keep a separate unknown-anomaly category.

Evidence to include in the final submission

Submit feature definitions, units, example measurements and class-wise observations.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	

Corrective action	To be completed during test	

TECHNICAL EXPANSION 08 – High-level defect decision and confidence management

Purpose. Design the decision stage that converts candidate regions into actionable defect events.

Engineering analysis

The high-level stage combines candidate features with recipe-specific decision rules. It should decide whether a region is a hole, broken thread, oil stain, weaving irregularity, colour variation or unknown anomaly. Rules should consider physical size, shape, contrast, texture and location. A single global intensity threshold is unlikely to work for all fabric types. The decision output should include class, confidence, bounding box, centroid, timestamp, encoder coordinate and a recommended action. Unknown or low-confidence cases should not be silently accepted as defect-free.

Worked calculation / design note

Define a defect event record containing a unique event ID, roll ID, fabric recipe, image reference, class, confidence, x/y position in millimetres, encoder count, processing time and marker acknowledgement. This schema separates image evidence from machine control and supports traceability.

Verification procedure

Replay saved images through the classifier and compare predicted labels with independent annotations. Measure confusion between stains and normal yarn shading, and between broken threads and ordinary weave texture.

Evidence to include in the final submission

Submit the decision flowchart, event schema and confusion matrix template.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 09 – CPU scheduling and 0.5-second response budget

Purpose. Allocate processing time across acquisition, preprocessing, analysis and communication on the available 4-core CPU.

Engineering analysis

The computer has four CPU cores, 16 GB RAM and no GPU. A compact, deterministic pipeline is more appropriate than an unbenchmarked deep network. Use bounded queues, avoid unnecessary image copies and process candidate regions rather than repeatedly processing the entire image. Acquisition and PLC communication should not be blocked by slow file writes. The decision deadline should be measured from image exposure or encoder trigger to the committed marker event, not only from the start of the classifier function.

Worked calculation / design note

An example engineering budget could reserve 50 ms for acquisition and transfer, 60 ms for preprocessing, 180 ms for feature extraction and decision, 40 ms for PLC communication and 70 ms for scheduling margin. The total is 400 ms, leaving 100 ms before the 0.5 s limit. These are proposed budget allocations, not measured timings; the actual values must be profiled on the production PC.

Verification procedure

Use time.perf_counter_ns() around each stage. Record median, 95th percentile and maximum latency during a sustained run. Confirm no queue growth and no dropped frames.

Evidence to include in the final submission

Submit a stage-wise latency table and CPU utilisation log.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 10 – Pipeline selection and maintainable software architecture

Purpose. Choose an explainable pipeline that can be maintained by the mill's engineering team.

Engineering analysis

A three-level pipeline is recommended as the main architecture: low-level image conditioning,

intermediate feature extraction and high-level defect decision. It gives more information than low-level processing alone and is easier to inspect than an opaque deep-learning-only design. A small machine-learning classifier may later be added to the high-level stage if labelled data justify it, but deterministic preprocessing and geometric calibration should remain explicit. Software should be modular so camera acquisition, image processing, event storage and PLC communication can be tested independently.

Worked calculation / design note

Separate modules can be named acquisition.py, calibration.py, preprocess.py, features.py, classifier.py, marker.py and logger.py. Define a stable data structure between modules so a camera change does not require rewriting the defect decision logic.

Verification procedure

Run unit tests on each module, integration tests with saved images and hardware-in-the-loop tests with the marker disabled initially. Record software version in each event.

Evidence to include in the final submission

Submit a software module diagram, test plan and version-control procedure.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 11 – RGB representation and colour-difference analysis

Purpose. Evaluate RGB data for colour variation and stain detection on cotton and blended fabric.

Engineering analysis

RGB stores red, green and blue channel values. It preserves the sensor's colour information and is useful when a colour variation changes one or more channels. It also provides a direct representation for display and export. However, RGB channel values are affected by illumination spectrum, exposure and white balance. A colour shift caused by lighting may resemble a dye variation. The inspection recipe should therefore use fixed camera settings and a reference patch or calibrated colour target.

Worked calculation / design note

For an 8-bit RGB image, each pixel stores $3 \times 8 = 24$ bits, or 3 bytes. A 2448×2048 image contains 5,013,504 pixels and requires approximately 15,040,512 bytes in uncompressed RGB, before temporary arrays and file headers. This is an approximate memory calculation.

Verification procedure

Capture a neutral reference and several known colour-variation samples. Compare raw RGB distance and a perceptual colour metric after calibration. Check centre and edge-of-field samples.

Evidence to include in the final submission

Submit a colour calibration record and channel comparison table.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 12 – HSV representation and stain discrimination

Purpose. Explain how hue, saturation and value can assist colour and oil-stain analysis.

Engineering analysis

HSV separates colour type, colour purity and brightness. Hue may help distinguish differently coloured yarns, while saturation can indicate whether a region is less or more chromatic. Value can highlight dark or bright stains. HSV is not immune to illumination changes: value changes with brightness, and hue becomes unstable when saturation is very low. For that reason, hue-based rules should be conditioned on saturation and should be validated on each fabric recipe.

Worked calculation / design note

A practical rule may first identify candidate stain pixels by local value difference, then use saturation or hue deviation as supporting evidence. The thresholds must be estimated from sample images. Do not interpret a fixed HSV threshold as universally valid for all dyes and blends.

Verification procedure

Compare RGB and HSV features on the same labelled samples. Record true detections, false alarms and the effect of changing illuminance. Inspect low-saturation grey and white yarn regions separately.

Evidence to include in the final submission

Submit an RGB-versus-HSV experiment sheet and selected recipe thresholds.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 13 – Grayscale representation and texture inspection

Purpose. Use grayscale data for structural defects where texture is more important than colour.

Engineering analysis

Grayscale reduces each pixel to one intensity value and is efficient for edge, line and texture processing. It is useful for finding holes, broken threads and local weave disturbances when those defects create a luminance difference. It cannot preserve colour distinctions that have similar luminance. Therefore, grayscale should be a processing branch rather than the only stored representation. For the low-contrast oil stain, stable exposure and sufficient bit depth are important because the specified contrast is only 5–8 grey levels.

Worked calculation / design note

At 8 bits, one grayscale pixel occupies one byte. The same 2448×2048 image requires approximately 5,013,504 bytes as an uncompressed grayscale array, one-third of the RGB pixel storage. This saving can reduce processing and memory bandwidth.

Verification procedure

Run edge and texture filters on grayscale crops of known broken threads and weave defects. Compare with colour-based features for oil stains and colour variations.

Evidence to include in the final submission

Submit a defect-to-channel selection matrix and example grayscale feature maps.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 14 – Multi-representation fusion and memory control

Purpose. Combine colour and grayscale information without overwhelming the CPU or RAM.

Engineering analysis

A practical design can acquire and retain the colour image, create a grayscale branch for texture, and create HSV features only for candidate regions. This avoids converting and storing multiple full-size arrays for every frame when they are not needed. The original colour image should remain linked to the event for audit. Candidate-region processing reduces computation because detailed colour statistics are calculated only where the initial detector finds an anomaly.

Worked calculation / design note

For a 5.01 MP image, one BGR array is about 15.0 MB and one grayscale array about 5.0 MB. Holding BGR, HSV and grayscale simultaneously can require about 35 MB before temporary buffers. Ten simultaneous frames can therefore exceed 350 MB. This is manageable in 16 GB RAM, but repeated copies and multiple worker queues can increase the footprint.

Verification procedure

Profile resident memory during a full production-length run. Check that memory use stabilises rather than increasing continuously. Confirm that saved crops preserve enough context to identify the fabric location.

Evidence to include in the final submission

Submit a memory budget and a channel-fusion flowchart.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 15 – Representation selection by defect class

Purpose. Set a defensible channel strategy for all five defect categories.

Engineering analysis

Use grayscale texture and edge features for holes, broken threads and weaving irregularities. Use HSV or calibrated RGB differences for colour variation and oil stains. Retain the original colour image for review. A stain may appear as a subtle brightness shift, but the specified 5–8 grey-level contrast makes lighting control and quantization equally important. The representation choice must be evaluated per fabric recipe because cotton and cotton-polyester surfaces can reflect light differently.

Worked calculation / design note

Build a matrix with defect classes as rows and candidate channels as columns. For each cell, record whether the channel gives strong, moderate or weak evidence based on sample measurements. Do not claim an experimental detection rate until tests have been completed.

Verification procedure

Validate the matrix with at least several independent examples of each defect type and include normal fabric variations that could cause false alarms.

Evidence to include in the final submission

Submit the channel-selection matrix and final processing recipe.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 16 – Four-connectivity and component separation

Purpose. Show how 4-connectivity groups pixels using only horizontal and vertical neighbours.

Engineering analysis

Four-connectivity uses the north, south, east and west neighbours. Diagonal foreground pixels are not directly connected. This can keep two corner-touching defects separate, but it may split a thin diagonal defect into several components. The number of foreground pixels does not change; only their grouping changes. Component count, centroid and bounding boxes can therefore differ depending on connectivity.

Worked calculation / design note

Consider a binary pattern with foreground pixels at (1,1), (1,2), (2,3) and (3,3). Under 4-connectivity, (1,1) and (1,2) form one component, while (2,3) and (3,3) form another. This example is illustrative and should be drawn in the report as a grid.

Verification procedure

Implement connected-component labelling under 4-connectivity and compare labels against a manually prepared expected map.

Evidence to include in the final submission

Submit the binary pattern, labelled output and component count.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 17 – Eight-connectivity and diagonal textile structures

Purpose. Evaluate the effect of including diagonal neighbours in defect-region labelling.

Engineering analysis

Eight-connectivity adds the four diagonal neighbours to the 4-neighbour set. A diagonal chain becomes a single component. This can be beneficial for irregular holes, broken fibres and ragged stains whose pixels touch at corners. It can also merge two separate regions that only touch diagonally. In textile imagery, diagonal yarn structures and irregular stain boundaries make this distinction important.

Worked calculation / design note

Using the illustrative pixels from the preceding pattern, 8-connectivity may join the diagonal pair at (1,2) and (2,3), resulting in one connected component if the coordinates are interpreted as adjacent diagonal locations. The exact result should be shown with a coordinate grid to avoid ambiguity.

Verification procedure

Create test masks containing diagonal lines, two corner-touching blobs and narrow bridges. Compare 4- and 8-connected component labels and review false merging.

Evidence to include in the final submission

Submit an 8-neighbour diagram and a component comparison table.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 18 – Mixed connectivity and ambiguity reduction

Purpose. Explain mixed connectivity as a way to avoid ambiguous diagonal connections.

Engineering analysis

Mixed or m-connectivity uses 4-neighbour links and allows diagonal links only when the common

4-neighbours do not create an alternate foreground connection. This definition helps prevent some ambiguous paths in binary images. It is conceptually useful when corner contacts could incorrectly merge two defects. Implementation requires care because standard connectedComponents calls commonly expose 4- or 8-connectivity directly; an m-connectivity labeller may need a custom neighbour rule or a carefully defined preprocessing step.

Worked calculation / design note

Test the rule on a 2×2 block with foreground pixels at opposite corners and on a pattern where one of the shared orthogonal neighbours is also foreground. Write down whether the diagonal link is permitted in each case before coding.

Verification procedure

Verify the custom labeller against hand-labelled small binary patterns. Include diagonal chains and loops so the rule is not tested on only one example.

Evidence to include in the final submission

Submit the m-connectivity rule, pseudocode and validation patterns.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 19 – Connected-component measurements and physical units

Purpose. Convert pixel components into meaningful defect measurements.

Engineering analysis

After labelling, measure pixel area, bounding box, centroid and orientation. Convert pixel coordinates to physical units using the calibration map. For a perspective-corrected plane, a constant millimetres-per-pixel scale may be used only if the calibration confirms it across the field. For nonuniform mapping, transform component boundary points into fabric coordinates before computing physical dimensions. Do not compare raw pixel area across different resolutions without conversion.

Worked calculation / design note

If horizontal pitch is 1200/2448 = 0.4902 mm/pixel and vertical pitch is 400/2048 = 0.1953 mm/pixel for the medium area-frame example, one pixel represents about 0.0957 mm². This demonstrates why a rectangular pixel grid can have different x and y scales.

Verification procedure

Use a calibrated grid to compare known physical shapes with measured component dimensions. Report measurement error and component merge/split events.

Evidence to include in the final submission

Submit a component measurement table with pixel and millimetre values.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 20 – Connectivity recommendation and defect-specific post-processing

Purpose. Select a connectivity policy that balances irregular-defect grouping and separate-region counting.

Engineering analysis

Eight-connectivity is a reasonable default for ragged defects because it preserves diagonal continuity. However, it can merge distinct corner-touching stains. A robust pipeline should inspect region shape, intensity distribution and narrow bridges, and should allow splitting when there is evidence of two separate centres. Four-connectivity can be retained as a diagnostic comparison. Mixed connectivity is useful when diagonal ambiguity is a recurring issue and its implementation has been validated.

Worked calculation / design note

Compare component count, area, centroid and bounding box using all three methods on a fixed set of masks. The preferred method should minimise both missed merges and false merges, not simply produce the fewest components.

Verification procedure

Run the comparison on holes, thread breaks, oil stains and weave irregularities separately. Review each error with the quality engineer.

Evidence to include in the final submission

Submit the connectivity decision table and representative annotated examples.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 21 – Rolling-shutter imaging under moving-web conditions

Purpose. Analyse the risk of rolling-shutter distortion at 0.6 m/s.

Engineering analysis

A rolling-shutter sensor exposes or reads different rows at different times. A moving fabric can therefore shift while the frame is being formed, producing skew or geometric distortion. The amount depends on the sensor's row-readout interval and exposure timing, neither of which is specified in the assignment. Rolling shutter may be inexpensive and adequate for slow or static inspection, but the line speed makes timing behaviour a critical selection criterion.

Worked calculation / design note

If the total row-readout interval is T seconds, the web displacement during readout is 0.6T metres. For example, if T were 10 ms, the web would move 0.006 m = 6 mm during the readout. This is an illustrative example only; use the selected camera datasheet for the actual T.

Verification procedure

Capture a moving grid or printed reference at 0.6 m/s. Measure skew and compare it with a stationary capture. Record the actual sensor readout timing.

Evidence to include in the final submission

Submit a rolling-shutter distortion test and camera timing evidence.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 22 – Global-shutter imaging and exposure control

Purpose. Explain why global shutter improves geometry but still requires short exposure.

Engineering analysis

A global-shutter sensor exposes all pixels at substantially the same time, reducing row-dependent geometric skew. It is suitable for area imaging of moving fabric when paired with a sufficiently short exposure. Global shutter does not eliminate motion blur: the web continues moving during the exposure interval. A short exposure, bright uniform illumination and stable mounting are therefore required. Triggering from an encoder can align images to physical web position.

Worked calculation / design note

Motion blur distance is v×t. At v = 0.6 m/s and t = 0.25 ms = 0.00025 s, blur = 0.6×0.00025 = 0.00015 m = 0.15 mm. This is one-tenth of the 1.5 mm minimum defect size and is a reasonable initial target for testing.

Verification procedure

Capture a moving edge target at several exposure times. Measure edge spread and verify that oil stains remain visible at the shorter exposure.

Evidence to include in the final submission

Submit the exposure-time experiment and selected shutter mode.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 23 – Line-scan camera geometry and encoder triggering

Purpose. Evaluate line-scan acquisition for continuous fabric-web inspection.

Engineering analysis

A line-scan camera captures one cross-web line at a time. The fabric's movement creates the second image dimension. This is naturally suited to continuous web inspection because the line can cover the 1,000 mm fabric width while the encoder controls sampling along the machine direction. The camera field of view is 1,200 mm, leaving nominal side margin around the 1,000 mm web. Correct alignment and calibration are required so that both fabric edges remain in the inspected region.

Worked calculation / design note

To obtain a 0.15 mm machine-direction pitch at 0.6 m/s, the line rate is 0.6/0.00015 = 4,000 lines/s. If a coarser 0.2 mm pitch is selected, line rate is 3,000 lines/s. These are design calculations; sensor line-rate capability must be checked.

Verification procedure

Use an encoder to trigger lines at fixed travel increments. Test at several speeds and verify that machine-direction scale remains constant.

Evidence to include in the final submission

Submit line-rate calculation, encoder setup and sample reconstructed image.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 24 – Optics, depth of field and lighting for moving fabric

Purpose. Specify optical and lighting checks that preserve the smallest defect.

Engineering analysis

Camera selection cannot be separated from lens selection. The lens must resolve the spatial frequency represented by the sensor and provide sufficient depth of field for fabric flutter. A 700 mm mounting height does not by itself determine magnification; focal length, sensor size and working distance are also required. Diffuse lighting reduces harsh reflections, while grazing or dark-field illumination can emphasise surface defects. Cotton and cotton-polyester may respond differently to the same lighting arrangement.

Worked calculation / design note

Keep the exposure short enough to limit blur, then adjust illumination to maintain a useful signal without clipping highlights. A polariser may reduce glare, but it can also reduce total light and must be tested against the low-contrast oil-stain requirement.

Verification procedure

Check focus across the full 1,200 mm field and at the expected range of fabric height. Inspect corners and centre with a calibrated target.

Evidence to include in the final submission

Submit lens specification, mounting drawing, lighting map and focus validation.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 25 – Camera selection and acquisition acceptance

Purpose. Conclude the acquisition comparison using the given speed, defect size and decision deadline.

Engineering analysis

A line-scan camera with encoder triggering is the preferred architecture for continuous web inspection. A global-shutter area camera is a viable alternative if the image overlap and data rate are manageable. A rolling-shutter camera is the least suitable because motion can introduce geometry errors. The 1.5 mm defect requirement means sampling, exposure and optics must be validated together. The 0.5 s decision requirement also requires a bounded pipeline and not merely a fast sensor.

Worked calculation / design note

At 0.6 m/s, the fabric moves 300 mm in 0.5 s. Since the marker is 300 mm downstream, the system must complete its decision within the travel time and account for marker actuation delay. A position-based trigger is more robust than an uncorrected timer.

Verification procedure

Perform a rated-speed acceptance run and report missed defects, line drops, latency percentiles, image scale and marker position error.

Evidence to include in the final submission

Submit camera selection justification and acquisition acceptance checklist.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 26 – Pixel-count comparison and spatial sampling

Purpose. Calculate pixel totals and physical sampling for all three candidate resolutions.

Engineering analysis

Total pixels equal image width multiplied by image height. The low-resolution option has

1280×1024 = 1,310,720 pixels. The medium option has 2448×2048 = 5,013,504 pixels. The high option has 4096×3000 = 12,288,000 pixels. With the full 1,200 mm field mapped to the horizontal pixel count, horizontal pitch is field width divided by image width. These values estimate sampling only; lens resolution and motion blur affect effective detail.

Worked calculation / design note

Low: 1200/1280 = 0.9375 mm/pixel; 1.5 mm spans 1.60 pixels. Medium: 1200/2448 = 0.4902 mm/pixel; 1.5 mm spans 3.06 pixels. High: 1200/4096 = 0.2930 mm/pixel; 1.5 mm spans 5.12 pixels. High resolution provides more samples across the defect but requires more bandwidth and CPU time.

Verification procedure

Use a calibrated target with known 1.5 mm features and compare visibility at centre and edges. Test at production speed rather than relying only on static images.

Evidence to include in the final submission

Submit the resolution calculation table and sample crops.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 27 – Two-dimensional sampling and field geometry

Purpose. Account for the vertical field dimension and unequal pixel pitch.

Engineering analysis

The assignment gives a 1,200×400 mm field of view. If the image width and height correspond to these dimensions, horizontal and vertical sampling must be calculated separately. The medium option gives 1200/2448 = 0.4902 mm/pixel horizontally and 400/2048 = 0.1953 mm/pixel vertically. This anisotropic sampling affects area, shape and orientation measurements. Algorithms should use physical calibration rather than assume square pixels in fabric coordinates.

Worked calculation / design note

Approximate pixel area for the medium example is 0.4902×0.1953 = 0.0957 mm². A 1.5 mm square defect would cover about 2.25/0.0957 = 23.5 pixels if fully resolved and aligned with the grid. Its width along the cross-web direction, however, is only about 3.06 pixels. This distinction shows why pixel count alone can be misleading.

Verification procedure

Capture a square calibration target and compare measured x and y dimensions. Apply geometric correction if the mapping is not uniform.

Evidence to include in the final submission

Submit separate x/y scale values and a physical-coordinate calibration record.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 28 – Sampling adequacy and the Nyquist limitation

Purpose. Explain why a defect represented by only a few pixels is difficult to classify reliably.

Engineering analysis

A feature that spans one or two pixels may be detectable as an intensity change but cannot be described reliably by its shape. At least several pixels across the smallest feature are desirable for stable boundary and texture analysis. The medium option gives about 3.06 pixels across 1.5 mm horizontally; the high option gives about 5.12 pixels. The low option gives only 1.6 pixels and is inadequate for reliable shape-based detection. The optical system must also provide adequate modulation transfer at the relevant spatial frequency.

Worked calculation / design note

Use the number of pixels across the defect as a first screening calculation, not a guarantee. A 1.5 mm defect rotated diagonally or partly hidden by yarn can occupy fewer effective pixels than its nominal width suggests.

Verification procedure

Test defects at multiple orientations and contrast levels. Record detection probability as a function of physical size.

Evidence to include in the final submission

Submit a sampling adequacy statement with limitations.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 29 – Image memory, bandwidth and storage trade-offs

Purpose. Compare uncompressed image data sizes and explain their impact on the CPU.

Engineering analysis

For an 8-bit grayscale image, memory per frame equals the pixel count in bytes. For 24-bit

RGB/BGR, multiply by three. Low resolution therefore requires about 1.31 MB grayscale or 3.93 MB colour per frame. Medium requires about 5.01 MB grayscale or 15.04 MB colour. High requires about 12.29 MB grayscale or 36.86 MB colour. These figures exclude compression, metadata and intermediate arrays. The high-resolution option can quickly dominate transfer and processing time.

Worked calculation / design note

At 10 colour frames per second, the medium option would generate approximately 150.4 MB/s before compression. This is a hypothetical frame rate used to illustrate the formula; the actual camera frame rate is not supplied. A line-scan design should instead calculate line bytes multiplied by line rate.

Verification procedure

Measure camera transfer rate and disk write speed. Test whether image acquisition continues while the processing and storage threads are active.

Evidence to include in the final submission

Submit a bandwidth budget and storage estimate based on the selected camera's real acquisition rate.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 30 – Resolution selection and validation plan

Purpose. Select a practical sampling option for the existing CPU and budget.

Engineering analysis

The medium resolution is a practical starting point for a CPU-only balanced configuration, but its

3.06 horizontal pixels across a 1.5 mm defect provide limited shape information. High resolution gives about 5.12 pixels across the defect and is more attractive for detection, but its data volume is about 2.45 times the medium pixel count. The final choice must be based on measured CPU latency, lens performance and the missed-defect target. If the 2% miss-rate requirement cannot be met at medium resolution, the design must move to a higher-resolution acquisition strategy or revise the budget.

Worked calculation / design note

Worked selection: compare pixel count, horizontal sampling, pixels across a 1.5 mm defect and measured CPU latency in one table. Select the lowest data rate that passes the defect-coupon test and the 2% missed-defect criterion. If no candidate passes both, state that a higher-resolution sensor or additional processing resources are required.

Verification procedure

Compare all candidate resolutions using the same fabric samples, illumination and defect annotations. Keep exposure and image processing parameters controlled. Report processing time and miss rate together rather than selecting resolution based only on visual sharpness.

Evidence to include in the final submission

Submit a resolution decision record and a test report with per-class recall.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 31 – Quantization levels and grey-level resolution

Purpose. Calculate the available intensity codes for the three quantization alternatives.

Engineering analysis

An n-bit quantizer provides 2^n intensity codes. Eight-bit quantization provides 256 levels, 12-bit provides 4096 levels and 4–6 bit quantization provides 16–64 levels. The 5–8 grey-level oil-stain contrast is small compared with the full 8-bit range. Reducing bit depth can collapse the difference into one or zero code steps, especially if the conversion is poorly scaled. Higher bit depth preserves more tonal detail but also increases storage and may not improve the image if sensor noise or illumination instability dominates.

Worked calculation / design note

At 6 bits mapped across 0–255, one code step is approximately 255/63 = 4.05 original grey levels. A 5-level contrast is only about 1.23 six-bit steps. At 4 bits, one step is 255/15 = 17 levels, so a 5–8 level stain difference may be lost. Eight-bit data preserve the contrast as 5–8 code levels if the sensor signal is stable.

Verification procedure

Capture a low-contrast reference target and compare the stain contrast after fixed 8-, 6- and 4-bit mappings. Preserve the original source values for audit.

Evidence to include in the final submission

Submit a quantization table and before/after contrast observations.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 32 – Quantization noise and sensor noise

Purpose. Distinguish quantization effects from sensor noise and lighting variation.

Engineering analysis

Quantization maps a continuous or high-resolution signal into discrete codes. Quantization error is only one part of the total image noise. Sensor read noise, photon noise, fixed-pattern noise, compression and illumination flicker can also obscure the stain. If the noise amplitude is comparable to 5 grey levels, retaining 12 bits alone will not guarantee reliable detection. Stable lighting, adequate exposure and flat-field correction are therefore essential. A 12-bit sensor can provide useful headroom, while an 8-bit processing branch may be adequate after calibration.

Worked calculation / design note

Measure the mean and standard deviation of a uniform fabric patch across repeated frames. Compare the stain-to-background difference with the background noise standard deviation. A contrast-to-noise ratio can be estimated as absolute mean difference divided by pooled standard deviation. Define an acceptance threshold from observed samples.

Verification procedure

Collect repeated images with unchanged fabric and lighting. Separate spatial variation from frame-to-frame variation and identify whether the dominant source is optical, sensor or quantization related.

Evidence to include in the final submission

Submit a noise measurement sheet and contrast-to-noise analysis.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 33 – Exposure, dynamic range and low-contrast oil stains

Purpose. Explain why correct exposure is necessary to retain subtle stain information.

Engineering analysis

If bright yarn highlights saturate, differences in those regions are clipped. If the exposure is too low, sensor noise can dominate the small stain signal. The camera should be set so the normal fabric occupies a useful portion of the sensor range without clipping. The 500–3,000 lux ambient variation is large enough that uncontrolled room light could change the apparent grey levels. An enclosure and regulated LED illumination are preferable to automatic exposure that changes unpredictably from frame to frame.

Worked calculation / design note

Use a reference grey patch and sample images at low, nominal and high illumination. Record histogram clipping at both ends and the stain contrast in the local region. Choose the exposure and light level that preserve stain contrast while avoiding saturation.

Verification procedure

Repeat the test on both cotton and cotton-polyester, because their reflectance and surface texture may differ. Check at the centre and both edges of the 1,200 mm field.

Evidence to include in the final submission

Submit exposure settings, histogram summaries and a stain visibility report.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 34 – Bit-depth processing strategy and data handling

Purpose. Define where high-bit-depth data should be preserved and where 8-bit conversion is acceptable.

Engineering analysis

If the selected camera outputs 12-bit data, preserve native data for calibration and critical low-contrast analysis where storage permits. Convert to 8-bit only with a fixed, documented mapping for algorithms that do not need the extra precision. Do not apply independent per-frame auto-normalisation to classification inputs because it makes pixel values inconsistent across time. For display, contrast stretching may be used, but the original image must remain available for traceability.

Worked calculation / design note

Compare file sizes for 8-bit and 12-bit storage. A 12-bit value is commonly packed into 16-bit storage in memory, so the practical byte count may be two bytes per pixel rather than 1.5 bytes. Confirm the camera SDK's packing format before estimating storage.

Verification procedure

Measure actual file size and CPU time for the native format, an 8-bit processing copy and the compressed archive format. Verify that compression does not erase the subtle stain.

Evidence to include in the final submission

Submit a bit-depth policy and a data-retention format specification.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 35 – Quantization recommendation and acceptance criteria

Purpose. Choose a quantization approach suitable for 5–8 grey-level stain contrast.

Engineering analysis

Eight-bit quantization is the recommended minimum processing representation for this scenario. A 12-bit sensor or acquisition path is desirable if available within budget, particularly for low-contrast stains, but must be paired with stable illumination and a suitable exposure. Reduced 4–6 bit quantization is not recommended for the main defect detector because its step size can be comparable to or larger than the stain contrast. The selected approach should be tested on real stained fabric rather than accepted solely from theoretical code counts.

Worked calculation / design note

Define acceptance as a measured stain recall and false-alarm rate at the low-contrast end of the sample range. The system must also meet the overall maximum missed-defect rate of 2%, with a separate report for oil stains.

Verification procedure

Use a blinded test set with known stain locations and compare native, 8-bit and reduced-bit pipelines. Re-run after lighting or camera setting changes.

Evidence to include in the final submission

Submit the quantization decision and class-specific acceptance results.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 36 – Pinhole image formation and mounting geometry

Purpose. Explain how the camera forms an image of the fabric plane from a 700 mm mounting height.

Engineering analysis

A camera lens projects light from the fabric onto the sensor. In the ideal pinhole model, image coordinates depend on focal length and the ratio of object coordinates to depth. The 700 mm mounting height establishes the approximate working distance but does not determine the field of view by itself. Focal length, sensor dimensions, lens distortion and camera tilt are also required. The fabric should be treated as a plane for planar calibration, while flutter and wrinkles are sources of measurement error.

Worked calculation / design note

Use the ideal relation image scale proportional to focal length divided by object distance. Do not calculate a focal length without the sensor size and field-of-view geometry. The supplied 1,200×400 mm field should be verified physically with a calibration target.

Verification procedure

Place a known grid or checkerboard in the fabric plane and capture it across the field. Estimate intrinsic camera parameters and lens distortion using OpenCV calibration functions.

Evidence to include in the final submission

Submit camera geometry sketch, lens data and calibration images.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 37 – Perspective distortion and width measurement

Purpose. Show how camera tilt and perspective can affect the ±2 mm width requirement.

Engineering analysis

If the camera is not perpendicular to the fabric plane, points at different positions can have different image scales. Lens distortion can also bend straight fabric edges. Measuring width directly in raw pixels and applying one global scale can therefore introduce error. A planar homography can map image coordinates to a calibrated fabric plane. The edge detector should fit the fabric boundaries robustly and ignore lint, frayed fibres and shadows.

Worked calculation / design note

The allowed width error is ±2 mm on a nominal 1,000 mm fabric. This corresponds to 0.2% of nominal width. The complete measurement chain—camera calibration, edge localisation, fabric alignment and mechanical stability—must fit within this tolerance.

Verification procedure

Measure a certified 1,000 mm reference at several lateral positions and compare the mapped width. Repeat after camera remounting and at different fabric tensions.

Evidence to include in the final submission

Submit a width measurement error table and calibration acceptance criteria.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 38 – Camera calibration and planar correction

Purpose. Design a repeatable calibration process for physical coordinate measurement.

Engineering analysis

Calibration estimates camera intrinsics, lens distortion and the mapping from image pixels to the fabric plane. OpenCV can estimate camera matrix and distortion coefficients from multiple views of a known checkerboard. A homography can then map the plane to a top-down coordinate system. Calibration should be performed with the target at the same plane as the fabric. It cannot fully correct fabric height variation or stretch, so mechanical control remains important.

Worked calculation / design note

Store calibration parameters with a version, date, camera serial number, lens focus and working distance. Recompute the mapping if the camera or lens is moved. Use a reference grid to check whether the calibrated width and position remain stable.

Verification procedure

Test at least the centre and four corners of the field. Report residual reprojection error and physical measurement error separately.

Evidence to include in the final submission

Submit the calibration procedure and a record template.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 39 – Illumination variation and controlled optical environment

Purpose. Analyse the 500–3,000 lux ambient range and propose a stable illumination arrangement.

Engineering analysis

The ambient range varies by a factor of six. Such variation can alter brightness and contrast,

especially for an oil stain with only 5–8 grey levels of difference. A shrouded inspection area with regulated LED lighting reduces the influence of factory lighting. Diffuse lighting supports uniform appearance, while dark-field lighting may reveal surface discontinuities. Cross-polarisation can reduce glare on reflective fibres, but reduces available light and must be tested with short exposures.

Worked calculation / design note

Measure illuminance at multiple positions across the 1,200 mm field. Create a flat-field reference image and calculate the residual variation after correction. Use the same camera exposure and gain across the test sequence.

Verification procedure

Test the five defect classes under the minimum and maximum expected ambient conditions. Compare detection and false-alarm rates with and without the light enclosure.

Evidence to include in the final submission

Submit an illumination layout, lux map and flat-field correction report.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 40 – Width-measurement algorithm and uncertainty budget

Purpose. Build a measurement chain that can demonstrate ±2 mm accuracy.

Engineering analysis

Detect the left and right fabric edges, map their points into physical coordinates, fit robust edge lines and calculate the perpendicular distance between them. Exclude local frays and isolated lint from the fitted edge. Report the width together with a confidence or quality flag. An uncertainty budget should include pixel localisation, calibration residual, lens distortion, camera alignment, fabric flutter and edge ambiguity. If the combined uncertainty approaches 2 mm, the design requires improvement before production acceptance.

Worked calculation / design note

Use the uncertainty budget as a structured sum of independent contributions where justified. Do not assume all errors are independent; systematic scale error can affect every measurement in the same direction. A conservative worst-case sum may be appropriate for initial acceptance.

Verification procedure

Measure a certified width standard repeatedly at different field positions and fabric speeds. Report mean error, standard deviation and maximum absolute error.

Evidence to include in the final submission

Submit the width algorithm flowchart and uncertainty budget.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 41 – Complete inspection-cell architecture

Purpose. Integrate camera, illumination, processing, storage, network and marker into one industrial system.

Engineering analysis

The camera acquires the moving web while regulated illumination provides stable contrast. The PC performs calibration, preprocessing, defect analysis and width measurement. Defect events and selected image evidence are stored locally. Gigabit Ethernet carries event messages to a PLC or marking controller. The marker is located 0.3 m downstream and uses encoder position to act on the correct portion of fabric. A feedback sensor should confirm that a mark was made. The system should include alarms for camera, light, encoder, storage and network failures.

Worked calculation / design note

Use the data path: camera/encoder → acquisition buffer → image processing → defect event → position scheduler → PLC → marking mechanism. In parallel, store the image reference and event metadata. Keep the real-time marker path independent of slow archive writes.

Verification procedure

Perform an interface test with the marker disabled, then a dry-run with a simulated encoder coordinate, and finally a controlled production trial.

Evidence to include in the final submission

Submit a complete block diagram, I/O list and failure-response table.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 42 – Budget configuration: minimum-cost trade-off

Purpose. Evaluate a low-cost design and explain why it may fail critical requirements.

Engineering analysis

A budget configuration may use a low-cost area camera, basic LED lighting and CPU processing. It can be suitable for a demonstration or preliminary defect screening, but limited resolution, uncontrolled illumination and rolling-shutter effects may make small defects unreliable. The low-resolution option represents a 1.5 mm defect with only about 1.6 horizontal pixels across the field, which is inadequate for dependable shape-based analysis. Low-cost equipment should not be selected solely because it fits a purchase limit.

Worked calculation / design note

Estimate the total cost by itemising camera, lens, light, mount, enclosure, encoder, PLC interface, marker, cables and installation. Use supplier quotations; do not treat illustrative prices as confirmed procurement values. Include contingency for mechanical integration and commissioning.

Verification procedure

Benchmark the budget configuration with the smallest defect coupons and low-contrast oil stains. Record which requirements fail and the cost of corrective upgrades.

Evidence to include in the final submission

Submit an itemised bill of materials and a risk-based suitability assessment.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 43 – Balanced configuration: CPU-centred industrial design

Purpose. Develop a balanced configuration that uses the available PC and meets the project budget.

Engineering analysis

The balanced design should prioritise a suitable industrial camera, controlled lighting, encoder triggering and an efficient CPU pipeline. The existing 4-core PC and 16 GB RAM are retained. A medium or high sampling option should be selected based on actual measured latency and recall. Use line-scan acquisition if the line is continuous and camera bandwidth supports it. Store defect crops and metadata rather than every full-resolution frame unless storage calculations show that seven-day full retention is feasible.

Worked calculation / design note

The total project budget is ₹1,50,000. The existing PC and 2 TB storage are stated as available system resources; the cost plan must clearly state whether they are excluded from the purchase budget. If they must be purchased, the allocation changes materially. Obtain quotations for all components and reserve contingency.

Verification procedure

Measure the 95th percentile and maximum latency on the exact PC. Run at rated conveyor speed and verify the 2% miss-rate requirement on a representative test set.

Evidence to include in the final submission

Submit the balanced bill of materials, cost assumptions and performance evidence.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 44 – Premium configuration: performance benefits and cost risk

Purpose. Compare a premium camera and processing platform against the budget ceiling.

Engineering analysis

A premium configuration could use a high-resolution industrial line-scan camera, high-output regulated illumination, dedicated acquisition hardware and a more powerful processing computer. It may improve spatial sampling and throughput, but it is likely to exceed the ₹1,50,000 limit unless some equipment is already available or the supplier offers a package. Higher resolution also increases image bandwidth and storage, so the processing architecture must scale with the sensor. A premium camera connected to an inadequate PC can simply move the bottleneck downstream.

Worked calculation / design note

Compare total cost of ownership, not only camera price. Include lens, light, mounting, encoder, interface card, cables, marker integration, software development, maintenance and training. State which costs are estimates and which are vendor-confirmed.

Verification procedure

Request quotations and test a representative data stream on the target processing platform. Measure end-to-end latency and storage throughput.

Evidence to include in the final submission

Submit a premium-versus-balanced comparison and a procurement risk note.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 45 – Marker position prediction and PLC communication

Purpose. Calculate how a defect location is translated into a physical mark 0.3 m downstream.

Engineering analysis

The system should convert each defect centroid into a cross-web coordinate and an encoder position along the fabric. The marker command is scheduled for the encoder count corresponding to the defect's machine-direction position plus the camera-to-marker distance. This is more reliable than a fixed delay because conveyor speed can vary. The PLC message should include event ID, x-coordinate, target encoder count, class and acknowledgement status. The marker should confirm actuation where possible.

Worked calculation / design note

At 0.6 m/s, travel time over 0.3 m is 0.3/0.6 = 0.5 s. At a different speed v, travel time is 0.3/v. If the decision takes the full 0.5 s, no time remains for actuator delay. Therefore, actual decision time should be lower than 0.5 s or the marker should be moved farther downstream.

Verification procedure

Test marker offset using a printed reference target. Measure the difference between detected defect coordinate and physical mark. Repeat at several speeds.

Evidence to include in the final submission

Submit the PLC message format, encoder calculation and marker offset test.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 46 – Seven-day image retention and system acceptance

Purpose. Plan image retention and define evidence for final acceptance.

Engineering analysis

Seven-day retention must be calculated from the actual image acquisition rate, pixel format and compression ratio. A 2 TB drive does not guarantee seven-day retention without knowing the data rate. A practical strategy stores full context around defects and a lower-rate audit sample of normal fabric, with metadata for every event. If full continuous recording is required, the storage budget must be calculated before installation. The system should monitor free space and delete only data older than the retention period.

Worked calculation / design note

Seven days contain 604,800 seconds. Required storage equals average bytes per second multiplied by 604,800. Include filesystem overhead and a safety margin. Do not use a guessed compression ratio; measure it on representative fabric images.

Verification procedure

Run a storage soak test and verify file retrieval, event-image linkage and automatic retention. Acceptance also includes width error, defect recall, latency and marker confirmation.

Evidence to include in the final submission

Submit a seven-day storage model and final commissioning checklist.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 47 – Changed export-order requirements and system gap analysis

Purpose. Identify the impact of the new 0.4 mm defect, UV stain, doubled throughput and faster line.

Engineering analysis

The new order changes the spatial, spectral, temporal and throughput requirements simultaneously. The current sampling must be reassessed because a 0.4 mm defect is much smaller than the original 1.5 mm target. The current visible camera may not detect a UV-responsive stain if its sensor and optics reject the relevant wavelength. Conveyor speed rises from 0.6 to 1.2 m/s, doubling motion displacement for a fixed exposure. The decision limit falls from 0.5 to 0.3 s while production volume doubles. The existing PC may become a throughput bottleneck.

Worked calculation / design note

At 1.2 m/s, the web travels 360 mm in 0.3 s. The marker is only 300 mm downstream, so a 0.3 s decision is later than the defect's nominal arrival at the marker by 60 mm of travel. The geometry or decision time must change; software alone cannot overcome this physical timing mismatch.

Verification procedure

Create a requirements gap table and classify each gap as optical, mechanical, computational, storage or control-related.

Evidence to include in the final submission

Submit a gap analysis and a list of non-negotiable design changes.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 48 – Sampling requirement for a 0.4 mm defect

Purpose. Calculate the image sampling needed to represent the new smallest defect.

Engineering analysis

A 0.4 mm feature needs substantially finer sampling than the original 1.5 mm feature. If a minimum of five pixels across the defect is used as a design target, horizontal pitch should be no greater than 0.4/5 = 0.08 mm/pixel. Across a 1,200 mm field, this implies at least 1,200/0.08 = 15,000 pixels across the web. This is a demanding sensor requirement and may require multiple cameras or a different optical arrangement. Five pixels is an engineering target for shape stability, not a universal law.

Worked calculation / design note

Compare the required cross-web pixel count with the existing camera candidates. The high option at 4096 pixels gives 1200/4096 = 0.293 mm/pixel, so a 0.4 mm defect spans only about 1.37 pixels. This is not adequate for reliable shape classification.

Verification procedure

Verify sampling with calibrated 0.4 mm defect coupons at several orientations and at production speed.

Evidence to include in the final submission

Submit the pixel-pitch calculation and a camera architecture proposal.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 49 – Motion, exposure and doubled production throughput

Purpose. Analyse the acquisition and processing consequences of doubling web speed and volume.

Engineering analysis

At 1.2 m/s, motion blur doubles for the same exposure. To maintain a 0.15 mm blur distance,

exposure must be 0.15/1200 = 0.000125 s = 0.125 ms. Shorter exposure reduces received light and makes the illumination system more demanding. If inspection volume doubles, the acquisition and processing data rate may also double. The PC must be benchmarked at the new input rate; if the queue grows, the system cannot satisfy the real-time requirement.

Worked calculation / design note

Line rate for a 0.08 mm machine-direction pitch at 1.2 m/s is 1.2/0.00008 = 15,000 lines/s. This is an illustrative target if equal sampling is required in the travel direction. The chosen sensor and interface must sustain this rate with adequate bit depth.

Verification procedure

Test sustained throughput with the actual image format. Measure dropped lines, queue depth, CPU utilisation, memory bandwidth and end-to-end latency.

Evidence to include in the final submission

Submit the new line-rate and exposure budget.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 50 – UV illumination and additional imaging channel

Purpose. Explain the optical changes required to identify stains visible under ultraviolet illumination.

Engineering analysis

UV inspection requires a source at a suitable wavelength and a sensor/optical path with adequate response at that wavelength. Many visible cameras include filters that reduce UV transmission. A dedicated UV-sensitive camera, suitable lens and filter may be required. The UV source must be enclosed and interlocked to protect operators. The stain's excitation and emission characteristics should be obtained from sample testing; the exact wavelength cannot be selected from the assignment alone.

Worked calculation / design note

An additional channel may operate in parallel or in a time-multiplexed sequence. Parallel imaging increases data rate and alignment complexity. Time multiplexing reduces simultaneous data volume but may be unsuitable at high speed unless the exposure sequence is synchronised to the encoder.

Verification procedure

Test representative stained and unstained fabric under candidate UV wavelengths with supplier safety guidance. Confirm that the UV channel does not create unacceptable fabric heating or degradation.

Evidence to include in the final submission

Submit a UV optical specification, safety assessment and channel-registration plan.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 51 – Adaptation strategy comparison and unchanged budget

Purpose. Compare software-only, selective-upgrade and architecture-redesign strategies.

Engineering analysis

Software-only adaptation is the lowest-cost option but cannot recover spatial information absent from the current sensor, create UV sensitivity in an unsuitable camera or remove a physical marker-distance conflict. Selective upgrades may improve optics, lighting, encoder and camera capability, but the existing PC and budget may still limit throughput. A redesigned architecture using multi-camera or high-rate line-scan acquisition and dedicated processing is technically stronger but likely to exceed the unchanged budget or require staged deployment. The practical recommendation is to upgrade only what can be validated and explicitly negotiate any requirement that cannot be met within current constraints.

Worked calculation / design note

Use a feasibility matrix with criteria: 0.4 mm detection, UV stain detection, 1.2 m/s acquisition, doubled throughput, 0.3 s decision, budget fit and implementation risk. Mark a criterion as unmet if no physical or measured evidence supports it.

Verification procedure

Obtain camera and UV component quotations and benchmark the existing PC using a representative data stream. Do not approve production until all critical constraints are demonstrated.

Evidence to include in the final submission

Submit the strategy matrix and a management decision note.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 52 – Revised architecture, physical constraints and final recommendation

Purpose. Define the updated system and clearly identify requirements that need additional resources.

Engineering analysis

The revised system may use a high-resolution line-scan camera or multiple cameras, encoder-synchronised acquisition, regulated visible and UV illumination, an upgraded optical path, and hardware-assisted preprocessing. The PC can remain for supervisory control and logging, but it may not be able to process the full high-rate stream within 0.3 s. The marker must be moved farther downstream, its actuator latency reduced, or the decision time made shorter than the 0.25 s travel time to the current marker at 1.2 m/s. The unchanged budget creates a feasibility conflict that must be reported honestly.

Worked calculation / design note

At 1.2 m/s, travel time to the current marker is $0.3/1.2 = 0.25$ s. A 0.3 s decision deadline is physically too slow for that location. The defect travels $1.2 \times 0.3 = 0.36$ m before the decision, which is 0.06 m beyond the marker location. A later marker or faster decision is mandatory.

Verification procedure

Run a design review with production, quality, electrical and safety teams. Confirm the camera sampling, UV spectral response, line rate, PC throughput, marker distance and budget before committing to the export order.

Evidence to include in the final submission

Submit the updated block diagram, open-risk list and final feasibility conclusion.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 53 – APPENDIX K – SAMPLE DEFECT ANNOTATION STANDARD

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

Every defect annotation should include a unique image ID, roll ID, fabric type, defect class, bounding box, centroid, estimated physical dimensions, severity and annotator identity. The annotation guide must define the boundary convention: include the visible defect only, or include a surrounding context margin. Use the same convention for every annotator. Ambiguous examples should be labelled uncertain rather than forced into a class. Maintain an unknown-anomaly label for defects not covered by the current taxonomy.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Create a small labelled set containing holes, broken threads, oil stains, weaving irregularities, colour variations and normal fabric. Have a second reviewer independently label a subset. Calculate agreement and discuss disagreement before using the data to tune thresholds.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 54 – APPENDIX L – DATASET SPLITTING AND LEAKAGE CONTROL

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

Training, tuning and test data should be separated by roll or production batch wherever possible. Randomly splitting adjacent frames from the same roll can leak nearly identical texture into both training and test sets and produce an overly optimistic result. Reserve a completely unseen roll or fabric batch for final evaluation. Include both cotton and cotton-polyester in each relevant evaluation group, and report performance separately by recipe.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Record dataset version, image IDs, roll IDs, capture settings, class counts and split policy. Never move a difficult test image into the tuning set without documenting that the test set has changed.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 55 – APPENDIX M – DEFECT SEVERITY AND REWORK POLICY

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

Detection and severity are different decisions. A small hole may require rejection, while a minor colour variation may be accepted under a customer tolerance. The vision system should report measured properties and let a quality-approved rule map them to rework, mark or reject actions. Severity thresholds must be defined with the mill and customer specification. The system must not invent acceptance standards from image appearance alone.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Prepare a severity matrix with class, physical size, location, customer tolerance, required action and approval authority. Mark missing policy values as pending quality approval.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 56 – APPENDIX N – CALIBRATION DRIFT AND DAILY REFERENCE CHECK

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

Camera movement, lens focus change, vibration and mechanical maintenance can invalidate calibration. A daily reference check should use a known grid or width standard. Compare measured positions with the baseline and define a warning band below the final ±2 mm acceptance limit. If the check fails, prevent automatic width reporting from being treated as valid until recalibration is completed.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Maintain calibration version, check date, operator, reference artifact ID, measured error and corrective action. Keep historical results to identify gradual drift.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 57 – APPENDIX O – LIGHTING UNIFORMITY AND FLICKER TEST

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

Uniform lighting is necessary because local brightness changes can resemble stains. Measure illumination across the complete camera field and at the fabric plane. Test for temporal flicker by capturing a stationary reference over time at the intended exposure. The lighting driver should be stable and synchronised where necessary. A flat-field correction can compensate spatial shading but should not be used to hide unstable illumination.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Record minimum, maximum and average illuminance, field uniformity, frame-to-frame intensity variation and any visible flicker. Repeat at the factory's expected ambient-light extremes.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 58 – APPENDIX P – PROCESSING LATENCY AND QUEUE TEST

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

End-to-end latency includes exposure, camera readout, transfer, preprocessing, feature extraction, classification, event writing, network transmission and PLC acknowledgement. Measure each stage independently and also measure total time using a shared monotonic clock. The system should use bounded queues and report overload rather than silently dropping old images. Average latency alone is insufficient; 95th percentile and maximum latency should be reviewed.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Run a sustained test at the highest expected data rate. Record CPU utilisation per core, memory use, queue depth, dropped frames, disk throughput and event latency. Repeat after enabling image retention and network communication.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 59 – APPENDIX Q – MARKER REGISTRATION AND POSITION ERROR

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

The marker must place a visible mark at the correct cross-web and machine-direction location. Cross-web position depends on camera calibration and actuator alignment. Machine-direction position depends on encoder scale, camera-to-marker distance, PLC scheduling and actuator delay. A fixed time delay is vulnerable to speed variation. Encoder-position scheduling is recommended, with a feedback sensor where possible.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Use a test strip with known reference points. Measure mark offset in millimetres at several speeds and across the full web width. Store calibration offsets and verify them after mechanical adjustment.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 60 – APPENDIX R – STORAGE CAPACITY AND RETENTION AUDIT

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

Storage planning must use measured data rate. Record the number of images or lines per second, bytes per image or line, compression ratio and fraction of images retained. Seven days equal 604,800 seconds. Add a safety margin for metadata, filesystem overhead and production variation. The retention service should preserve images associated with defect events and maintain an auditable deletion log.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Perform a retention simulation using a representative production data stream. Confirm that old normal images are removed according to policy while defect evidence and event metadata remain available for the full required period.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 61 – APPENDIX S – NETWORK AND PLC FAULT RECOVERY

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

Gigabit Ethernet provides a communication interface but does not guarantee deterministic delivery. Use sequence numbers, event identifiers, acknowledgement and timeout handling. If the PLC connection fails, queue events locally and raise a visible alarm. The fail-safe response—stop the line, continue with manual inspection or divert fabric—must be selected by production management. Never silently discard a marker command.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Disconnect the network during a controlled test. Confirm that the fault is detected, event data are preserved, the line response follows the approved procedure and queued events are reconciled after reconnection.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 62 – APPENDIX T – SOFTWARE TESTING AND VERSION CONTROL

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

Use unit tests for calibration transforms, physical-unit conversion, component measurements and marker coordinate calculations. Use saved images for repeatable integration tests. Hardware-in-the-loop tests should begin with the marker disabled. Version-control the source code, configuration files and calibration data. Each production event should record the software and recipe version so that a result can be reconstructed later.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Maintain a release checklist covering code review, regression tests, performance test, configuration backup, rollback plan and operator notification.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 63 – APPENDIX U – OPERATOR TRAINING AND SHIFT HANDOVER

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

Operators should understand the meaning of defect classes, confidence flags, marker status, width warnings, camera/lighting alarms and review actions. Shift handover should include unresolved review items, calibration check status, storage capacity, marker faults and fabric recipe changes. Training should explain that the system is a decision-support tool with defined limitations, not a guarantee that every unknown defect is detected.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Prepare a one-page shift handover record with date, shift, roll ID, recipe, unresolved events, system alarms, calibration status, storage status and supervisor sign-off.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	

TECHNICAL EXPANSION 64 – APPENDIX V – FINAL ACCEPTANCE AND CUSTOMER TRACEABILITY

Purpose. Provide a practical engineering procedure that supports design, validation, commissioning and traceability.

Engineering analysis

Final acceptance should verify the full system under representative operating conditions. Critical criteria include the maximum 2% missed-defect rate, ±2 mm width accuracy, 0.5 s decision time, seven-day image retention and successful marker registration. Report each defect class separately and document any excluded condition. For customer traceability, each roll should have a summary showing inspected length, width statistics, defect count by class, marked locations and unresolved events.

Worked calculation / design note

Use the numerical values and constraints from the industrial scenario. Where a supplier parameter, physical measurement or customer tolerance is not provided, mark it as an assumption or pending confirmation. Do not present illustrative calculations as measured factory results.

Verification procedure

Use a signed acceptance sheet. Include test method, sample size, measured result, pass/fail status, evidence reference, responsible person and corrective action for every criterion.

Evidence to include in the final submission

Complete the associated record with actual test evidence, responsible person, date, result and corrective action.

Parameter / evidence	Value to record	Remarks / status
Fabric / roll ID	To be completed during test	
Camera / recipe setting	To be completed during test	
Measured result	To be completed during test	
Acceptance criterion	To be completed during test	
Corrective action	To be completed during test	